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The X-Minute City Paradox in São Paulo: local accessibility from everyday travel origins

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Abstract

Urban areas in the Global South face social inequality and climate change simultaneously. Disadvantaged residents usually reach essential services through long journeys made by the most polluting transport modes, making it urgent to develop solutions that narrow the accessibility gap while reducing carbon emissions. The 15-Minute City, and its X-Minute City generalisation, propose proximity as a response, yet the direct transfer of this model to unequal cities remains contested, calling for closer examination of how proximity is measured and which conditions produce it. Existing applications in the Global South measure local accessibility on foot alone, retain a single threshold, and rarely identify who benefits from nearby amenities. Drawing on the 2023 São Paulo Origin Destination Survey, this study asks how potential local accessibility varies across São Paulo's municipality, considering different transport modes and time thresholds; and how it varies across economic classes. It proposes the Local Accessibility Index (LAI), which brings together three time thresholds (15/20/30 minutes), different transport modes, and a carbon weighting to analyse local accessibility both spatially and individually. Place-based analysis revealed that local accessibility concentrates in São Paulo's centre and along rail corridors, with coldspots persisting in the outskirts across every amenity category and time threshold, and that accessibility falls with economic class. Cycling extends access beyond walking, while adding public transport provides little additional access beyond what walking and cycling already reach. Removing the carbon weight reduces the class gap. Individual-based analysis showed that inequality is greatest at home and smaller at work/study anchors, and that lower-income groups live where accessibility is lowest. Future work could explore how to make cycling more affordable, safe and inclusive, incorporate amenity quality into the index, and compare potential accessibility with real travel behaviour.

Declaration of Authorship

I, Beatriz Costa Werneck, declare that this dissertation is all my own original work, and that all sources have been acknowledged. Artificial Intelligence (Claude Code | Opus 5) was used for proof reading, grammar and code validation. It is 11,897 words in length, excluding figures, bibliography, footnotes and appendices.

Table of Contents

Abstract **2**

Declaration of Authorship **3**

Table of Contents **4**

List of Figures **6**

List of Tables **7**

Abbreviations **8**

Acknowledgements **9**

1. Introduction **10**

2. Study Area: São Paulo **12**

2.1.São Paulo's transport and socio-spatial context 13

2.2.Existing Policies in São Paulo 14

3. Literature Review **16**

3.1.Accessibility, proximity and the 15mC ideal 16

3.2.Global South and the 15mC ideal 17

4. Methodology **19**

4.1.Accessibility Index (LAI): conceptual basis and rationale 20

4.2.Data 21

4.2.1.Data inputs 21

4.2.2.Data filtering 22

4.2.3.Ethical Considerations 24

4.3.The index 25

4.3.1.Time thresholds 25

4.3.2.Amenity categories 25

4.3.3.Transport modes routing 25

4.3.4.Low-carbon principle 26

4.3.5.Computing the index 27

4.4.Hexagonal spatial representation 29

4.5.LAI interpretation 29

4.5.1.Place-based analysis 29

4.5.2.Individual-based analysis 30

5. Results **31**

5.1.Place-based analysis 31

5.1.1.Overall accessibility across São Paulo 31

5.1.2.Accessibility by amenity category 32

5.1.3.Accessibility across economic classes 34
5.1.4.Transport modes and carbon weight....36
5.2.Individual-based analysis....37
6. Discussion 41
6.1.Proximity spatial distribution....41
6.2.Reaching amenities: walking, cycling and public transport....42
6.3.Carbon emissions associated to proximity....42
6.4.For whom proximity is possible 43
6.5.Strengths and limitations 44
7. Conclusion....45
References 47
Appendix A. Supplementary data....51
Appendix B. Data availability 54
Appendix C. Research Log 55

List of Figures

Figure 01: Urban proportion in Global South countries 12
Figure 02: Mean public transport trip's duration starting at home, by origin zone 14
Figure 03: Methodology flowchart.....19
Figure 04: 2023 SP OD Survey zones 23
Figure 05: Origin points filtered from 2023 SP OD Survey.....24
Figure 06: Index flowchart.....27
Figure 07: LAI at X minutes 31
Figure 08: LAI's Getis-Ord Gi* analysis at X minutes 32
Figure 09: LAI per category at 15 minutes 33
Figure 10: LAI's Getis-Ord Gi* analysis at 15 minutes per category 34
Figure 11: Boxplot LAI per economic class at 15 minutes.....35
Figure 12: LAI per economic class at 15 minutes 35
Figure 13: Individual LAI per economic class at 15 minutes.....38
Figure 14: Anchors' LAI per economic class at 15 minutes.....39
Figure 15: Anchors' LAI and population density per economic class at 15 minutes 40

List of Tables

Table 01: Global Moran's I values of the per-category LAI at 15 minutes.....33
Table 02: Contribution of transport modes (carbon weighted).....36
Table 03: Contribution of transport modes (without carbon weight).....37
Table 04: Mean LAI per economic class with and without carbon weight.....37
Table 05: Modal share of trips, 2023 SP OD Survey (weighted by the trip expansion factor).....51
Table 06: Transport mode share within each economic class, 2023 SP OD Survey.....52
Table 07: Amenity categories and sub-categories OSM key and values.....52
Table 08: Research Log.....55

Abbreviations

Abbreviation	Full term
15mC	15-Minute City
2023 SP OD	2023 São Paulo Origin and Destination survey
ABEP	Associação Brasileira de Empresas de Pesquisa
CNEFE	National Cadastre of Addresses for Statistical Purposes
CPTM	Companhia Paulista de Trens Metropolitanos (Author's translation: São Paulo Metropolitan Railways Company)
GTFS	General Transit Feed Specification
OSM	OpenStreetMap
PDE	Plano Diretor Estratégico (Author's translation: Strategic Master Plan)
PlanClima	Plano de Ação Climática (Author's translation: Climate Action Plan)
PlanMob	Plano de Mobilidade (Author's translation: Mobility Master Plan)
POI	Point of interest
SDG	Sustainable Development Goals
SP	São Paulo
SRTM	Shuttle Radar Topography Mission
UN	United Nations
US	United States of America
XmC	X-Minute City
LAI	Local Accessibility Index

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1. Introduction

Urban areas face two of today's most complex and interlinked challenges: social inequality and climate change. In cities, different population groups experience distinct levels of access to services and resources. For the most disadvantaged, reaching a service that is not available locally often requires long journeys by the most polluting modes of transport. This problem is sharper in the Global South, where inequalities are higher and low-income groups tend to live in peripheral and under-serviced areas. Developing solutions that improve the gap in accessibility while reducing carbon emissions has therefore become urgent.

Accessibility is defined the ease of accessing activities and destinations, enabled by existing land use and transport system (Geurs and van Wee, 2004). Local accessibility, here understood as the ease of reaching opportunities nearby the community (Handy, 1993), has gained attention as a way of reducing dependence on motorised transport, increasing individuals' exposure to opportunities and improving their quality of life.

It is in this context that the 15-Minute City (15mC) concept emerged, in which residents everyday's needs should lie within a 15-minute walk or cycle from home (Moreno, 2016). The framework was revised by Moreno and colleagues, who established that the threshold should be adapted to each city's local context (Moreno et al., 2021), giving rise to the X-Minute City (XmC) concept (Logan et al., 2022). Yet a paradox remains: the belief that the 15mC and the XmC can serve as a simple, universal solution for sustainability and quality of life overlooks that, in Global South cities such as São Paulo, proximity is not always a reliable indicator of accessibility (Boisjoly et al., 2020). In these contexts, proximity often emerges from socio-economic exclusion and restricted mobility, while essential services remain further away (Nieto-Combariza et al., 2026). This challenges the direct transfer of the model, and calls for a closer look at how proximity is measured and which conditions produce it.

With this in mind, other studies proposed adaptations of the XmC framework, mostly regarding how much travel time is allowed (Victoria. Department of Environment, Land, Water and Planning, 2017; Greater Sydney Commission, 2018; Capasso Da Silva et al., 2019; Levinson, 2019), which transport modes are considered (Calafiore et al., 2022; Guzman, Oviedo and Cantillo-Garcia, 2024; Birkenfeld et al., 2024), and whether amenity diversity alone is sufficient to meet the community's needs (Mouratidis, 2024).

These variations matter because the XmC framework is increasingly mobilised to deliver global sustainability agendas, most directly the United Nations Sustainable Development Goals (SDGs) 11 and 13 (UN, 2015). SDG 11[1] calls for access to basic services and sustainable transport for all by 2030, while SDG 13[2] calls for climate measures to be integrated into urban planning policies.

[1] SDG 11: "Sustainable Cities and Communities"

[2] SDG 13: "Take Urgent Action to Combat Climate Change and its impacts"

This study addresses that gap by investigating inequalities in local accessibility in São Paulo, Brazil, based on the 2023 São Paulo Origin Destination Survey (2023 SP OD). A Local Accessibility Index (LAI) is proposed to compute potential accessibility for travel origin points from the 2023 SP OD at three time thresholds (15/20/30 minutes), across walking, cycling and public transport. The index considers a carbon weight for each transport mode, prioritises amenity variety, and enables results to be disaggregated by economic class. The results of LAI are used to answer the two research questions guiding this study:

RQ1) From origin points, how does potential local accessibility vary across São Paulo's territory, considering different transport modes and time thresholds?

RQ2) How does this potential accessibility from origin points vary across different economic classes in São Paulo?

The study makes methodological and empirical contributions. Methodologically, it proposes a replicable, data-based way to measure and diagnose sustainable local accessibility in Global South cities. Empirically, it reveals how potential local accessibility in São Paulo varies by mode, threshold, amenity type and economic class.

The dissertation is organised as follows. Section 2 introduces São Paulo as the study area, its transport and socio-spatial context, and existing policies related to local accessibility. Section 3 reviews the literature on the 15mC and XmC and on accessibility in the Global South. Section 4 details the methodology, including data wrangling and construction of the index. Section 5 presents the results across the territory, transport modes, amenity categories and economic classes. Section 6 discusses these findings in light of São Paulo’s known transport and inequality dynamics. Section 7 concludes by summarising the main findings and their implications for planning in São Paulo.

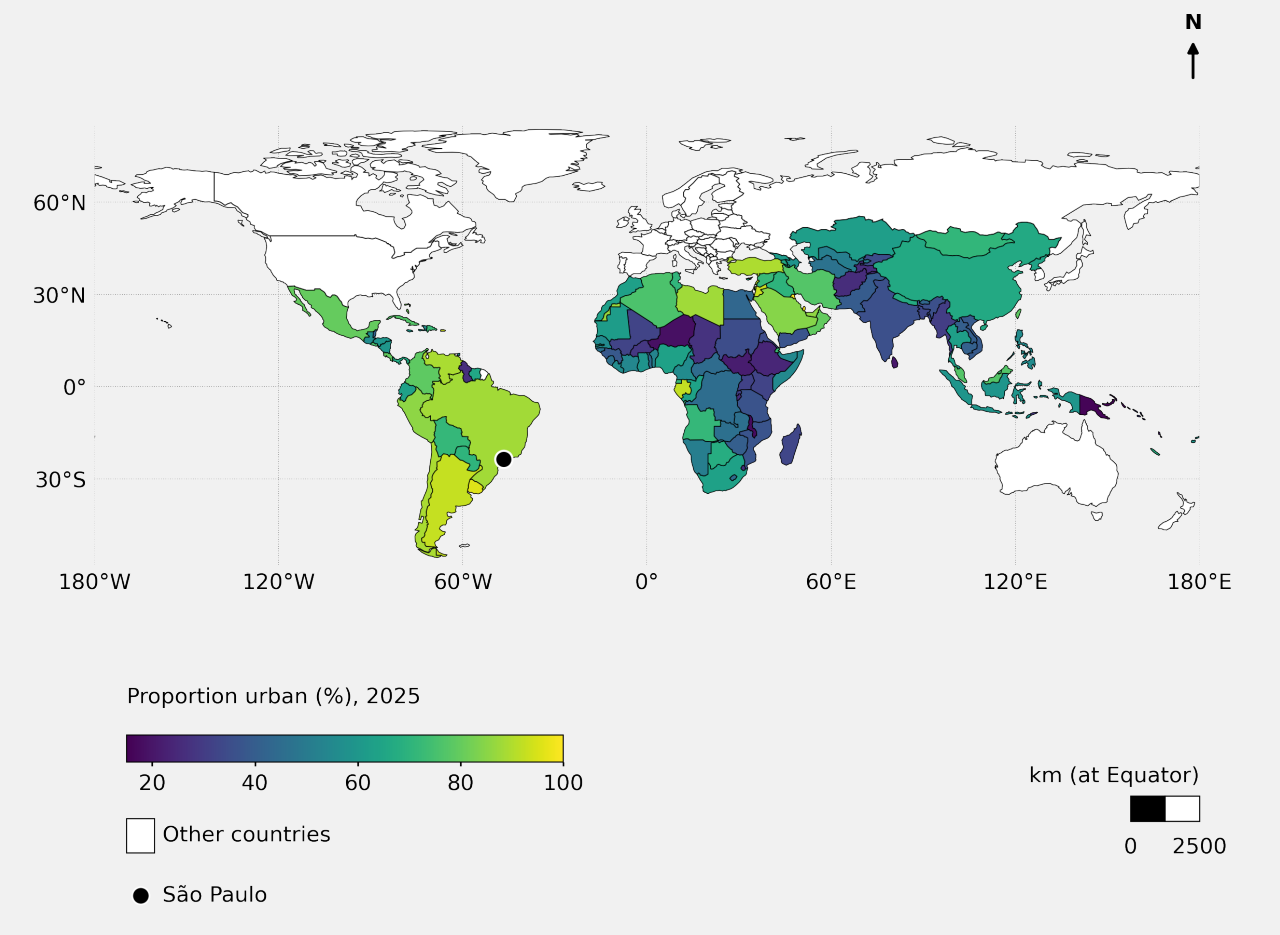
2. Study Area: São Paulo

Understanding what can be done to promote more accessible, inclusive and fair cities in the Global South is an urgent demand. Among Global South regions, Latin America and Caribbean is the one with the biggest urban proportion, in which South America stands out with 85.5% of its population in urban areas (UN, 2025). Figure 01 shows urban proportion in Global South countries.

Urbanisation in Latin America is entering a more stable phase, as the rapid expansion of previous decades is not projected to continue at the same rate (UN, 2025). Consequently, the successes and failures of Latin American cities might serve as a guideline for rising megacities in India, Nigeria and Pakistan, which according to the UN (2025) are expected to account for over half of global city growth by 2050.

The municipality of São Paulo is one of the 33 megacities in the world, and importantly, has an open and rich public dataset on transport and mobility: the 2023 SP Origin Destination (OD) Survey, carried out by the state-owned company Metrô.

Figure 01: Urban proportion in Global South countries



Source: Author (adapted from UN (2025) data)

2.1. São Paulo's transport and socio-spatial context

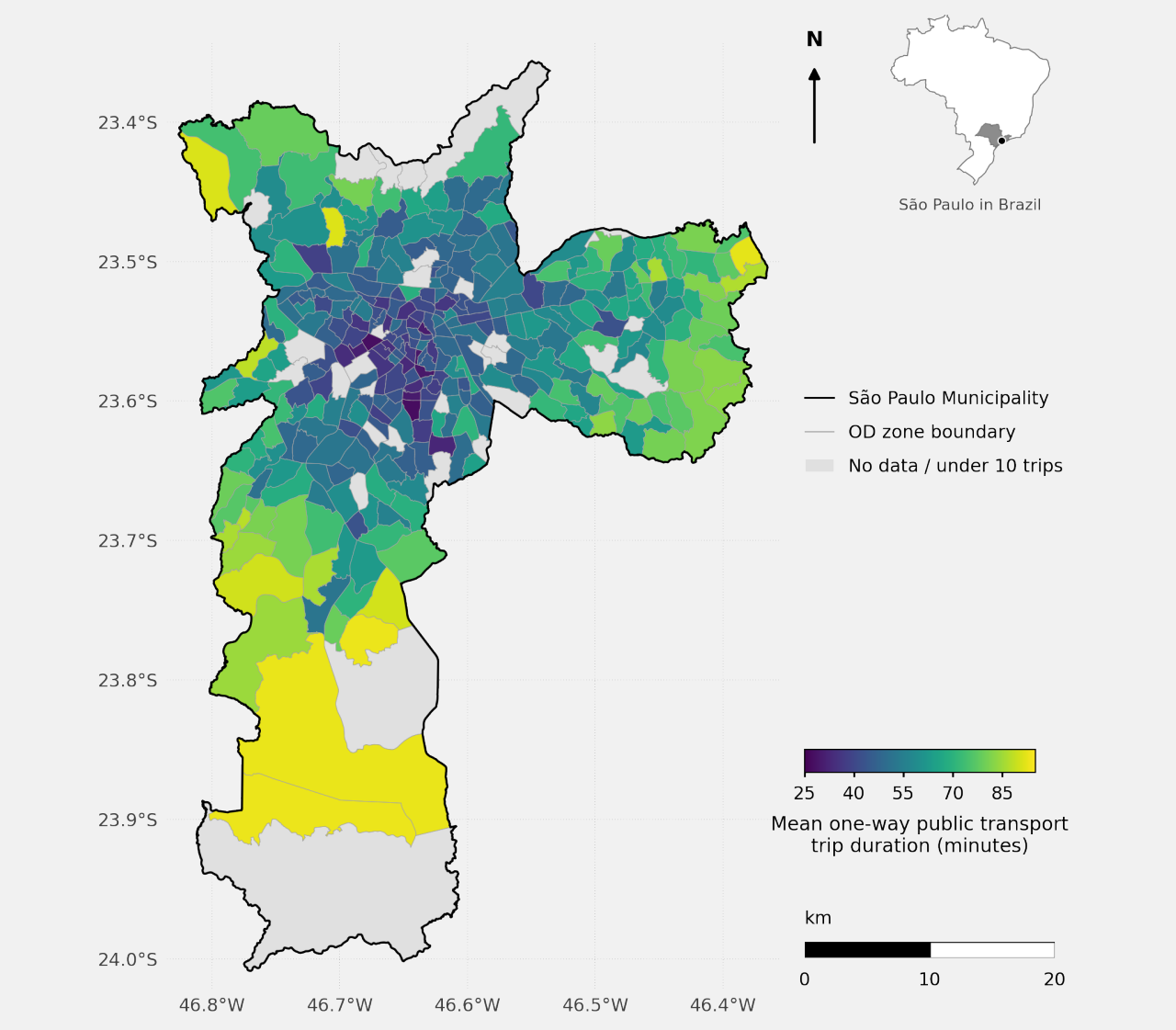
According to the 2023 SP OD Survey, 28.2% of journeys made on a typical weekday in São Paulo were on foot, 1.32% by bicycle, 0.19% classified as "other", 26.68% by public transport and 43.61% private motorised transport (Table 05 - Appendix A). However, these figures mask how mode shares differ per economic classes. Private car trip's share falls steeply as economic power decreases, from 66.4% among class A (richest) to 6.5% among class D–E (poorest). The opposite pattern is observed for the two most affordable modes: walking rises from 11.3% of trips in class A to 46.6% in class D–E, while public transport increases from 9.4% to 31.9% (Table 06 - Appendix A). As household income declines, trips shift away from the private car and go towards walking and public transport, underscoring the extent to which car use in São Paulo remains a privilege of the richest.

The average one-way public transport journeys, weighted by survey's trip expansion factor, took 66.1 minutes in 2023, more than four times the 15-minute 'ideal'. Furthermore, São Paulo's peripheral areas, recorded public transport journeys from home with average duration of 90 minutes, while the same measure in the city centre is around 30 minutes. Figure 02 shows the mean public-transport's duration from trips starting at home, by origin zone, weighted by trip expansion factor.

These journeys illustrate structural inequalities. São Paulo is marked by intense residential segregation, in which wealth, formal employment and urban infrastructure concentrate in the central areas, where the highest-income classes both live and work (Lisboa, 2022). The lower income group is mostly confined to distant outskirts, where the supply of jobs, services and quality public space is more limited, and from which reaching well serviced central areas demands longer and costlier journeys (Lisboa, 2022). Those least able to afford long commutes are therefore those who need to make them the most.

Accessibility research on São Paulo has measured this pattern directly. Vieira and Haddad (2015) built a gravity-based index of access to employment for the metropolitan region from the 2007 Origin Destination Survey, and found accessibility by private vehicle to be on average 22% higher than by public transport, with high-income and high-accessibility zones clustering together in central areas. Boisjoly et al. (2020) measured cumulative access to jobs within 60 minutes by public transport and found that 35% of the lowest-income households fall in the lowest accessibility quartile against 11% in the highest, while 56% of the highest-income households are in the highest quartile. Critically, they also showed that proximity to infrastructure is not the same as access, since even the lowest-income households living close to a rapid transit station remain well below the levels reached by wealthier households near the same stations. Working at municipal scale with a 90-minute post-work budget, Saraiva and Barros (2022) found that over 93% of individuals in classes D–E recorded accessibility below the city average, against more than half of classes A1 and A2 above it, with cyclists recording markedly higher accessibility than bus or public transport users.

Figure 02: Mean public transport trip's duration starting at home, by origin zone



Source: Author (adapted from 2023 SP OD Survey)

In São Paulo, accessibility is most found in the wealthy central areas, while peripheral areas have much less. There is therefore a mismatch between population needs and serviced areas. This uneven distribution is increased by the deep social inequalities that characterise many Global South cities, which make it harder to apply the 15mC concept fairly.

2.2. Existing Policies in São Paulo

São Paulo already has policy instruments that aimed at reducing the distances travelled in daily life and at shifting those journeys to active and public modes. The Strategic Master Plan[1] (São Paulo, 2014) explicitly seeks to bring housing and employment closer together, concentrating densification and mixed land use along public transport corridors in order to reduce the length and number of motorised journeys. The Municipal Urban Mobility Plan (São Paulo, 2015)

[1] Plano Diretor Estratégico, Law 16,050/2014, articles 13, 75–82 and 175; revised by Law 17,975/2023

prioritises walking, cycling and public transport over individual motorised modes, including targets for expanding the cycling network and improving pedestrian infrastructure. More recently, the Municipal Climate Action Plan (São Paulo, 2020) committed the city to carbon neutrality by 2050, with the transport sector as a central target.

These instruments would benefit from a city-wide, socio-economically disaggregated diagnostic of where the city already delivers local accessibility, by which modes, and for whom. The index proposed in this study meets this demand by identifying amenity-category gaps in the territory, quantifying the contribution of each low-carbon transport mode, and allowing progress to be monitored as new survey editions are released. In this sense, the research does not import a foreign ideal into São Paulo's planning framework, it offers a measurement tool for goals the city has already set itself.

3. Literature Review

This chapter discusses relevant existing literature that forms the basis for this study. It first addresses the concept of accessibility adopted in this study and provides a critical review of the 15mC concept (3.1). This is followed by discussion on the suitability and applicability of the 15mC concept for cities in the Global South (3.2).

3.1. Accessibility, proximity and the 15mC ideal

This study adopts Geurs and van Wee's (2004) definition of accessibility, in which accessibility is the product of the interaction between the spatial distribution of opportunities and the mobility available to reach them. They argue there are four interdependent components for accessibility: land-use, transport, temporal and individual.

The idea that high levels of accessibility at a local level lead to a better quality of life is not new. Perry (1929) proposed the neighbourhood-unit scheme as a basis for urban planning, in which a family's main destinations should be reached without crossing any highway. Jacobs (1961) argued that urban vitality depends on mixed uses and daily needs being met within walking distance. Handy (1993) defined local accessibility as the ease of reaching opportunities nearby the community. Hence, the 15mC ideal (Moreno, 2016) is aligned with previous critiques of functional zoning modernist planning, and its concept have four major components: proximity, diversity, density and ubiquity, the latter was subsequently replaced by digitalisation (Moreno et al., 2021).

The 15mC idea was quickly applied in policies. In 2020, Paris applied this concept in its mayoral programme through policies aimed at promoting local living and reducing car dependency (Moreno, 2021). Other proximity-based strategies followed in different cities of the Global North, such as Barcelona, through its superblocks model (Mueller et al., 2020), and Melbourne, whose metropolitan plan adopts "20-minute neighbourhoods" (Victoria. Department of Transport and Planning, 2017). The success of these initiatives is typically framed in terms of fewer and shorter car trips, higher shares of walking and cycling, reduced carbon emissions, and stronger local economies (Moreno et al., 2021; Allam et al., 2022).

Time threshold is perhaps one of the most debated aspects of the 15mC concept. Moreno (2016) proposed 15 minutes on foot or by bicycle as the ideal of proximity, but the model's authors later recognised that the threshold should be adapted to each city's morphology and context (Moreno et al., 2021). In practice, planning had already adapted it. Melbourne's metropolitan strategy adopted 20-minute neighbourhoods (Victoria. Department of Environment, Land, Water and Planning, 2017), a threshold that Capasso Da Silva et al. (2019) operationalised for the low-density, car-oriented context of Tempe, Arizona, while Sydney's plan embraced the 30-minute city (Greater Sydney Commission, 2018), which Levinson (2019) develops as a metropolitan-scale planning goal centred on access to jobs. Logan et al. (2022) formalise this flexibility as the "x-minute city", treating the threshold as an analytical parameter rather than a fixed rule.

Despite its rapid adoption, the 15mC has received significant criticism. Mouratidis (2024) identifies seven issues in the concept and its applications, four of which are directly relevant to this study: the focus on the quantity of destinations rather than their sufficiency, amenities aggregation, the disregard of public transport, and the assumption of a constant walk and cycle speed. Glaeser (2021) argues that the main function of cities is to connect people to jobs and opportunities across the urban area, and that planning for local self-sufficiency would mostly benefit residents of neighbourhoods that are already well served, warning that 15-Minute neighbourhoods might increase inequalities. Casarin, MacLeavy and Manley (2023) place the 15mC concept within a longer history of urban interventions that promised social mix and failed to deliver it, pointing to the risk of gentrification. Finally, Barthelemy (2026) shows that, because employment is concentrated in large firms, commuting times cannot fall below a certain limit regardless of planning decisions, which limits how far work can be incorporated into the living-locally ideal. Nieto-Combariza et al (2026) evaluate informal settlements where living locally naturally exists but accessibility is restricted to few mobility options and low-quality amenities, warning that proximity does not guarantee an improvement in quality of life. These critiques do not invalidate the 15mC ideal, but they indicate what an application of the XmC needs to be aware of.

Another debate concerns how the 15mC concept should be measured. Most assessments rely on place-based accessibility rather than on what people actually do (individual-based) (Zhang et al., 2025). An example of individual-based analysis was done by Zhang et al. (2025), in which they compared eighteen months of mobility traces in Finland against proximity-based expectations, and identified that residents of central areas, where amenities are dense, often travel further to reach specialised or higher-quality services. Conversely, people living on the edges of the city often stay in their own neighbourhoods mainly because they have fewer choices.

3.2. Global South and the 15mC ideal

Recent studies illustrate the growing interest in application of the 15mC concept in the Global South. In Lahore, Pakistan, a study used a GIS-based isochrone method to measure walking access to commerce, health, education and recreation across the city's street network, finding that only a fraction of the population could reach these amenities within 15 minutes (Baig et al., 2025). In Accra, Ghana, another study compared walking accessibility inside and outside privately master-planned "new cities" (e.g. Appolonia City), showing that fewer than half of metropolitan residents have 15-minute walking access to parks, schools or jobs, and that residents of the new purpose-built developments were in fact worse off than the rest of the city (Korah, Cobbinah, and Adu, 2024). In Bogotá, Colombia, a 15-minute proximity index based on the preferences stated by residents from different economic classes showed that the concept risks reinforcing existing socio-spatial inequality in developing cities characterised by segregation, unless it explicitly takes into account which economic classes would benefit from nearby amenities (Guzman, Oviedo and Cantillo-Garcia, 2024).

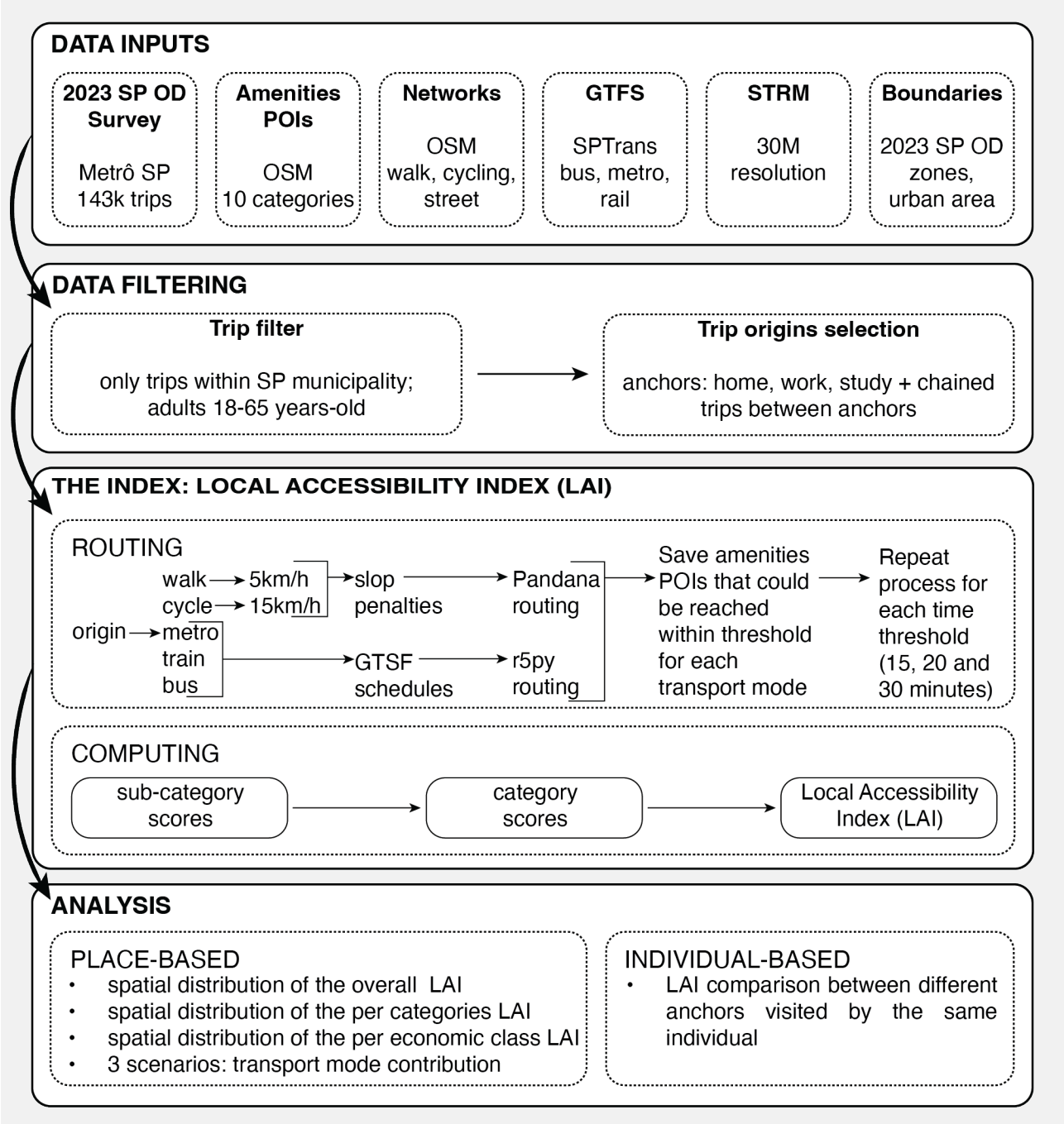
A fourth study, on Santiago de Chile, engages directly with the tension between proximity and segregation in Latin America (Israel et al., 2026). The authors measured the number of amenities per person within 500 metres of each city block. They found that middle-income neighbourhoods had better and more varied access than both richer and poorer ones. Access also remained high in the city centre, which was built up when wealthy residents still lived there and kept its amenities after those groups moved out, a common trajectory in Latin American cities. Crucially, they warn that in highly segregated cities an excessive focus on local accessibility could reinforce segregation, limiting opportunities for different groups to encounter one another (Israel et al., 2026). This warning aligns with evidence from the Global North: Zhang et al. (2025) found that increased localism improves social mixing only for central residents, while exacerbating experienced segregation elsewhere.

These studies show how the 15mC concept has been investigated in cities of the Global South. Access is measured by walking alone, or by a fixed-distance buffer in the case of Santiago. None includes cycling or public transport, despite their heavy use by low-income residents. The time threshold is rarely questioned, and population characteristics appear only in Bogotá's and Santiago's studies, each incorporating just one element at a time. None combines several transport modes, more than one time threshold and economic class in a single methodological framework, and the 15mC question has not been posed for São Paulo yet, despite the city having various accessibility studies.

4. Methodology

This chapter describes the data and methods used to build and analyse the Local Accessibility Index (LAI). Figure 03 summarises the full workflow.

Figure 03: Methodology flowchart



Source: Author

4.1. Accessibility Index (LAI): conceptual basis and rationale

Four factors shaped the design of the proposed index. They concern where accessibility is measured from, which amenities are counted, which transport modes are included, and how those modes are weighted in low-carbon emission perspective.

The first factor concerns where accessibility should be measured from. It is assumed that a person's access to amenities is not fixed to the home: it changes through the day depending on where the person is (Shrestha et al., 2019). This notion is often adopted in segregation studies, which understand that the segregation is experienced along the various locations that an individual frequents and spends meaningful time (Park and Kwan, 2018; Cai, Browning and Cagney, 2024). This rationale was brought here and applied to local accessibility. Hence, the proposed index measures accessibility from the various places which people spend time during the day.

To identify these places within the 2023 SP OD Survey, two criteria were used to filter and select the trip origins from which the local accessibility measure is calculated: anchors and stops made during chained trips. The anchors refers to the locations where people spend long, regular periods of time; defined here as home, work and study destinations. The stops made during chained trips capture the intermediate stops that occur between two anchors. Following McGuckin and Murakami (1999), a chained trip is understood as a sequence of linked journeys between two anchors, in which each stop becomes the starting point of the next. For example, dropping a child at school on the way from work to home. Applying both criteria therefore means that, for each individual, the index considers not only the residence but also the additional daytime origins from which accessibility is experienced.

A second factor is which amenities to count. The idea of a set of essential daily functions comes from the 15mC concept. Moreno (2016) states that residents should be able to reach six functions locally: living, working, commerce, healthcare, education and entertainment. However, not all amenity categories matter in the same way. Zhang et al. (2025) show that different types of amenity attract different travel behaviour, and warn that treating all categories as equally essential is not suitable. This connects to one of the criticisms discussed in Section 2.1, counting destinations without questioning whether the mix of destinations is sufficient (Mouratidis, 2024). For this reason, the proposed index looks at each amenity category separately and prioritises the variety of subcategories that can be reached, rather than to the total number of amenities within one category. The categories used in this study follow Zhang et al. (2025).

A third factor is which transport modes to include. The 15mC is based on walking and cycling only (Moreno, 2016), and most applications in the Global South measure accessibility on foot alone (Section 3.2). The disregard of public transport was identified as one of the main criticisms of the concept (Mouratidis, 2024). Public transport is included here, following Birkenfeld et al. (2024), since it still requires walking to and from stops and therefore remains consistent with the

active, low-carbon rationale, and since many residents of São Paulo depend on it to reach opportunities beyond their own neighbourhood.

A fourth factor is that transport modes should not be treated as equivalent. The rationale of the 15mC is to bring services closer and to reduce the carbon emissions of motorised travel (Moreno, 2016; Moreno et al., 2021; Allam et al., 2022). A fifteen-minute walk and a fifteen-minute metro trip both provide access, but they do not have the same environmental cost. This point is directly relevant to São Paulo: the city's Climate Action Plan identifies transport as its largest source of emissions, responsible for around two thirds of the total (São Paulo, 2020). To stay consistent with the low-carbon aim of the concept, the proposed index weights the contribution of each mode according to its life-cycle carbon emissions, so that access provided by low-emission modes counts for more.

Finally, the design of the index builds on existing work in São Paulo. Saraiva and Barros (2022) measured accessibility across the whole municipality using the 2017 São Paulo Origin Destination Survey, several transport modes and a hexagonal grid, showing that this kind of analysis is possible at the scale of the entire city. The present study places the accessibility analysis in São Paulo within the XmC debate and adds the following elements: multiple time thresholds, different transport modes, a carbon weight, and results disaggregated by economic class.

4.2. Data

4.2.1. Data inputs

The main dataset for this study is the anonymised 2023 São Paulo Origin and Destination Survey (2023 SP OD). This is a public dataset collected by Metrô, the state-owned company that plans, builds and operates the metro system in the São Paulo Metropolitan Region. The survey captures a full 24-hour day of travel, from 4:00 AM to 3:59 AM the following morning. Respondents were asked about trips from the previous workday, so the data reflect a typical weekday in the metropolitan area. However, one important limitation is that the sample excluded walk trips shorter than 500 metres unless the trip purpose was work or study.

In 2023, the survey validated information from 31,904 households, 79,331 people and 143,038 trips, of which 30,125 trips are records for people who made no trips. These figures were expanded to all metropolitan population using the survey’s expansion factors for trips, household, family, and person, calculated by Metrô proportionally to household, family and person control totals within each OD zone stratified by mean household income (Metrô-SP, 2023). When expanded by the survey's expansion factors, these numbers from the raw sample represent approximately 7.8 million households, 21.2 million people and 35.7 million done trips.

The 2023 SP OD dataset contains detailed information about each trip, including: the individual’s home and work/study location, household and individual economic characteristics, employment

and education status, the total number of trips each person made, origin and destination coordinates, trip purpose, transport mode, travel time, and an expansion factor.

For mapping points of interest (POIs), this study used data from OpenStreetMap (OSM), a public source widely used in spatial analysis, which allows amenities to be identified at a finer level of detail. The description of the used amenity categories and subcategories (OSM key values) are in Table 07 - Appendix A.

The walking and cycling networks for São Paulo were obtained from OSM as two separate mode-filtered extracts: a pedestrian network (foot-accessible ways), and a cycling network (cycleways and roads where cycling is permitted, excluding motorways). A third OSM extract, containing the full street network, was used to route the on-foot portions of public-transport trips. The public-transport stop coordinates, physical line geometry of each route, route stop sequences and timetables were taken from the GTFS feed published by SPTrans (SPTrans, 2026). This feed covered the municipal bus, metro and rail services.

Shuttle Radar Topography Mission (SRTM) elevation data were combined with the OSM walking and cycling networks for the impact of steep slopes on walking and cycling travel times.

A few additional spatial datasets supported the analysis. The municipal boundary of São Paulo, used to delimit the study area, clip the amenities and provide the base outline of the maps, was taken from the cartographic package that accompanies the 2023 SP OD survey. The same package supplied the survey's zones, which were used when georeferencing and processing the trip records. Finally, the urban area defined by São Paulo's 2014 Strategic Master Plan was added as a background layer to figures in this study. This layer was obtained from GeoSampa, the municipality's open spatial data portal (Prefeitura de São Paulo, 2014).

4.2.2. Data filtering

As the analysis was carried out for the municipality of SP, due to the unavailability of GTFS for the entire metropolitan region, only the trips that both begin and end within the São Paulo municipality were selected (Figure 04). Moreover, trips from users were filtered to include only adults aged 18–65, in order to capture a standard adult travel pattern and exclude children and older people, which have different travel behaviours.

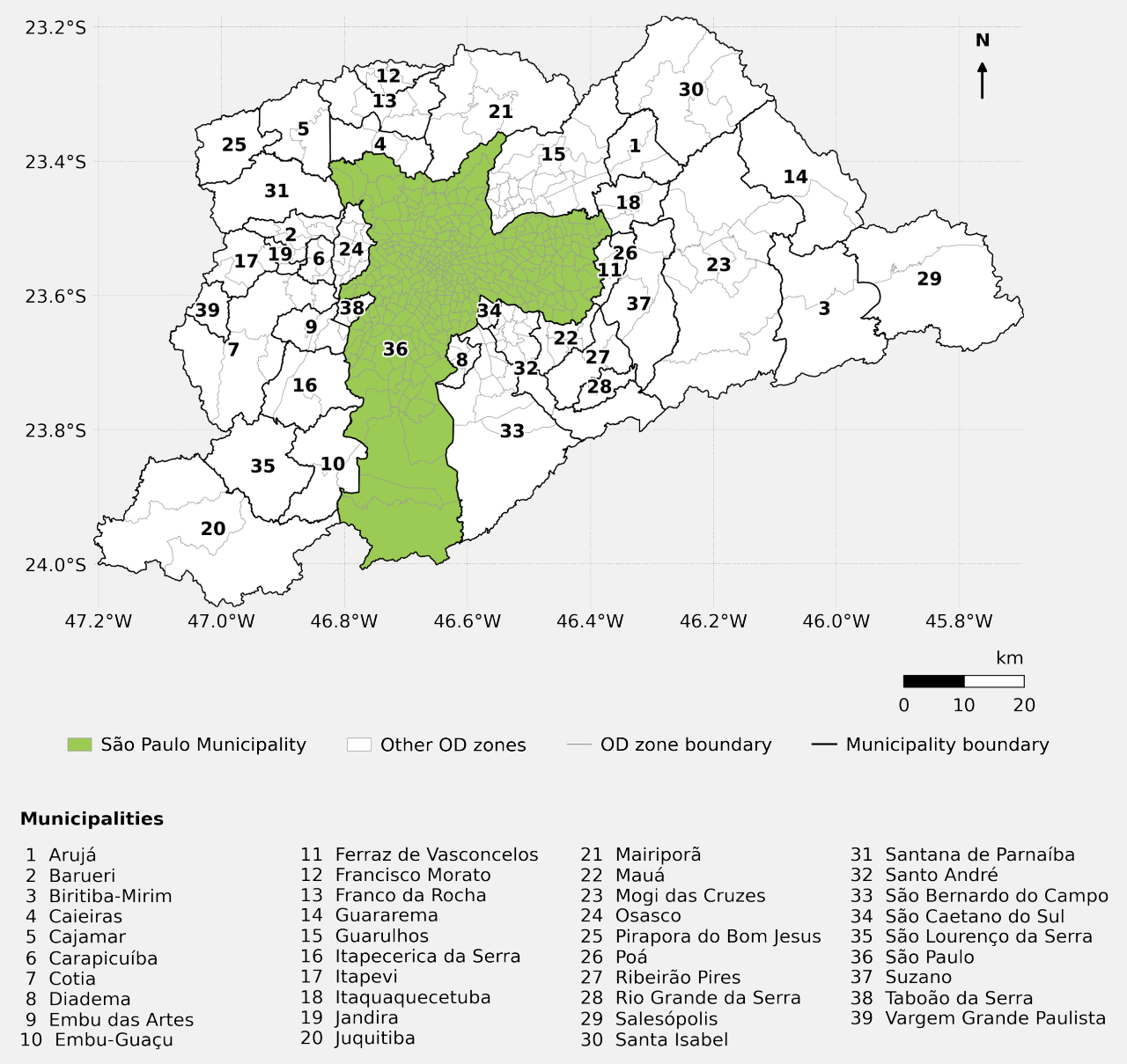
Following Saraiva and Barros (2022), each individual's economic profile was extracted from the 2023 SP OD survey, which classifies individuals into one of six economic classes (A, B1, B2, C1, C2 and D–E). This classification follows the Brazilian Economic Classification Criterion (CCEB) (ABEP, 2024), which estimates a household's purchasing power from the ownership of durable goods and the education of the head of the household.

In order to define the anchors that served as origins for this accessibility study, a trip-purpose filter was also applied to keep only origins that are related to long-stay activities. Furthermore, considering the 11 trip purposes at the origin in the 2023 São Paulo Origin Destination dataset,

this study selected only the ones related to residence, study and work (1-Work in industry, 2-Work in commerce, 3-Work in services, 4-School/Education and 8-Residence) as the ones of long duration.

The proposed index also calculated potential local accessibility for origins from chained trips. Following McGuckin and Murakami (1999) chained trips definition, each person's trips were ordered chronologically, and every origin that fell between two anchors was retained. Trips lying outside this anchored segment, namely those made before the day's first anchor or after its last, were not included.

Figure 04: 2023 SP OD Survey zones

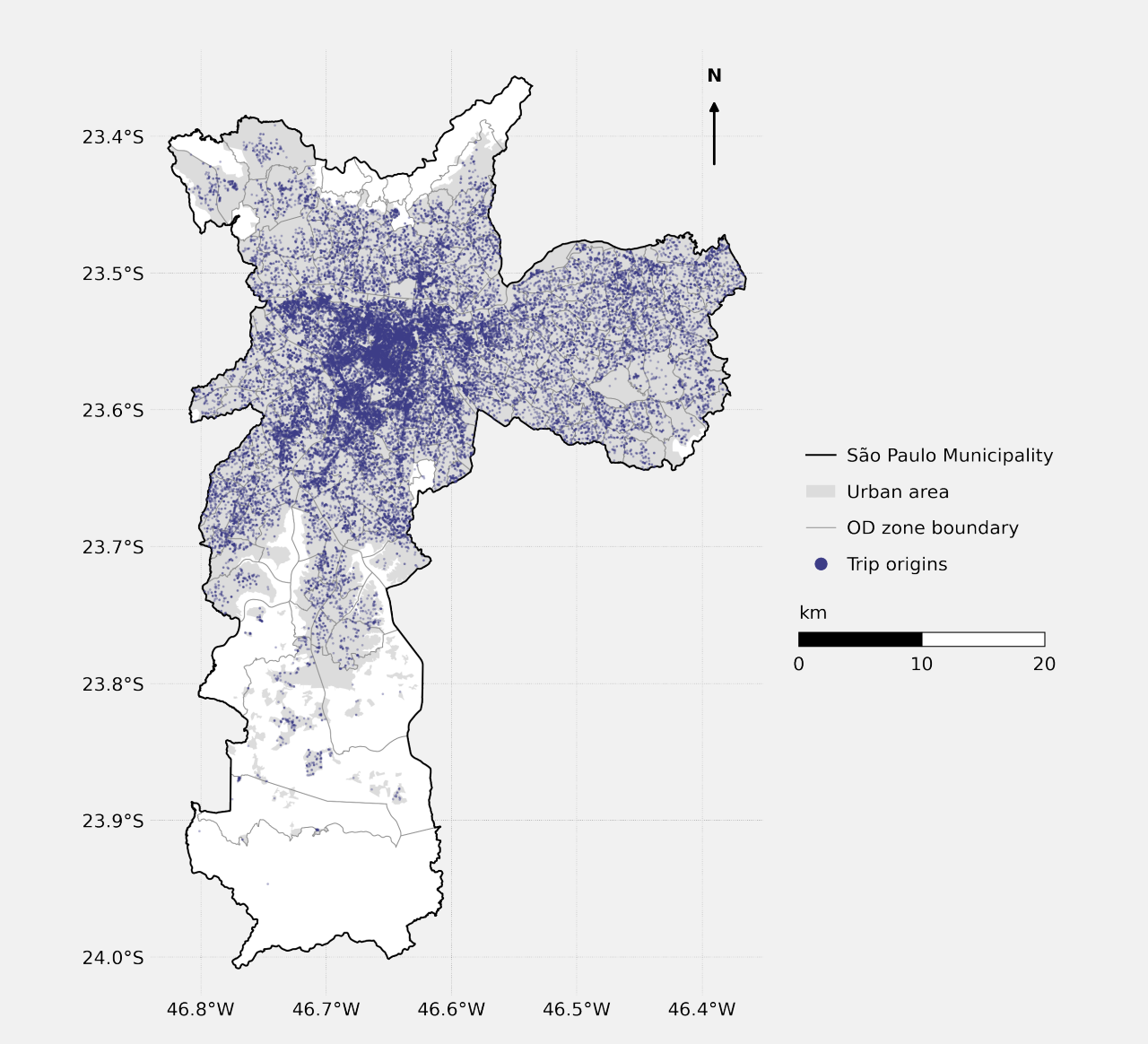


Source: Author

After filtering, the sample comprised 52,252 trips, made by 21,984 individuals from 14,876 households and distributed across 29,552 distinct origin locations (Figure 05). Weighted by the survey's expansion factors, these figures correspond to approximately 13.7 million trips

(13,724,447), 4.8 million people (4,817,931) and 3.0 million households (2,973,746). The resulting sample represents the typical weekday travel of São Paulo's adult population.

Figure 05: Origin points filtered from 2023 SP OD Survey



Source: Author

4.2.3.Ethical Considerations

This research uses only secondary, publicly accessible and anonymised datasets. All trip records in the 2023 SP OD Survey are anonymised and aggregated by Metrô prior to public release, and the analysis operates at the level of origin points and hexagonal spatial aggregations, with no attempt to identify, profile or reveal information about specific individuals or households. On this basis, the ethical risk of the research was classified by UCL as minimal/exempt. The research was therefore not subject to approval from UCL's Research Ethics Committee.

4.3. The index

This section introduces the Local Accessibility Index (LAI) and describes how this study used the selected data to propose an index that empirically measures the potential local accessibility of each of the filtered origins (coordinate points) from the 2023 SP OD survey. To this end, the index measures which types of amenities an individual could potentially reach from a given origin, considering different time thresholds and modes of transport.

The index is calculated for points where actual recorded trips originated and do not cover the entire territory as detailed in Section 4.2.2 (Figure 05). The advantage of this approach is that trips can be linked to individual characteristics of the person who carried out the trip, allowing for both place and individual-based analysis of results.

4.3.1. Time thresholds

Three time thresholds were selected: 15, 20 and 30 minutes. Treating the threshold as a parameter rather than a fixed rule follows the XmC concept itself, and each of the three values is already established in planning practice (Section 3.1). Furthermore, using more than one threshold avoids results that depend on a single choice of time, since accessibility changes with the threshold used (Pereira, 2019). Additionally, the upper limit of 30 minutes is set by the scope of this study, which measures local accessibility. Finally, for each threshold, the filtered origins were used to compute which amenities could be reached within X minutes (15, 20 and 30 minutes) on foot, by bicycle, and by public transport (bus, metro and train).

4.3.2. Amenity categories

This study adopted the ten amenity categories defined by Zhang et al. (2025): Civic & Religion, Culture, Dining, Education, Fitness, Groceries, Healthcare, Transport, Retail and Services. To capture internal diversity, each category was defined by a set of OSM tags, treated as sub-categories (Table 07 - Appendix A). Healthcare, for example, was divided into hospitals, clinics, doctors, dentists, pharmacies and health centres. What is recorded for each category is therefore how many of its sub-categories can be reached within the time threshold, rather than how many POIs of that category can be reached. This guarantees that the diversity of each amenity category is captured.

4.3.3. Transport modes routing

All routing and data processing were carried out in Python, with spatial data handled in GeoPandas and tabular data in Pandas.

Walking and cycling were routed with the Pandana library (Pandas Network Analysis), which returns the nearest-POI distances from each origin to each amenity across the mode's respective network. A walking speed of 5 km/h and a cycling speed of 15 km/h were adopted: 5 km/h falls

within the range of comfortable adult walking speeds reported by Bohannon (1997), and 15 km/h follows the Active Mode Appraisal Toolkit of the UK Department for Transport (2022).

Walking and cycling speeds were then adjusted for terrain using SRTM elevation data at 30-metre resolution (processed with the Rasterio and Elevation packages), from which the gradient of each network segment was derived. For walking, speed was scaled by Tobler's (1993) hiking function (gradient coefficient 3.5), which slows walking as the terrain steepens. For cycling, the effect of gradient on speed was taken from the regression of Parkin and Rotheram (2010) and applied to the adopted flat speed, reducing cycling speed by approximately 6.7% for each additional 1% of uphill gradient and increasing it by about 4.0% for each 1% of downhill. The travel time of each segment was then computed from its slope-adjusted speed.

Public transport travel times were routed using the R5 (Rapid Realistic Routing on Real-world and Reimagined networks) multimodal routing engine via the r5py Python package, rather than fixed average speeds, because door-to-door journeys depend on walking to stops, waiting time, in-vehicle time and transfers. To carry out the calculations, GTFS schedules for a typical weekday morning peak departure (7:30 am) were used. The morning peak was chosen because commuting to the main anchors of this study (work/study) concentrates in this period, and it is also when public transport runs at its highest frequency.

4.3.4. Low-carbon principle

To preserve the original 15mC low-carbon emphasis while still including public transport, each mode is assigned a weight inversely proportional to its life-cycle carbon emissions per passenger-kilometre. The operational emissions of each public-transport mode were taken from published inventories for São Paulo: 3.3 gCO_2e/pkm for rail (CPTM, 2024), 3.0 gCO_2e/pkm for the metro (Metrô-SP, 2022), and 90.6 gCO_2e/pkm for SPTrans buses (CPTM, 2024[1]). The embodied emissions of vehicle manufacture and supporting infrastructure were then added using the life-cycle multipliers of Chester and Horvath (2009). Bicycle embodied emissions were taken from the European Cyclists' Federation (2011). This gives life-cycle emissions of about 5 gCO_2e/pkm for cycling, 7.7 for metro, 8.3 for rail and 145 for bus. The weight of each mode is inversely proportional to these values (0.20 for cycling, 0.129 for metro, 0.120 for CPTM rail and 0.0068 for bus) so that lower-emission modes count for more. Walking is effectively carbon-free, so this rule would give it an unbounded weight. It is therefore set to the maximum of 1.00, the highest value any mode can take.

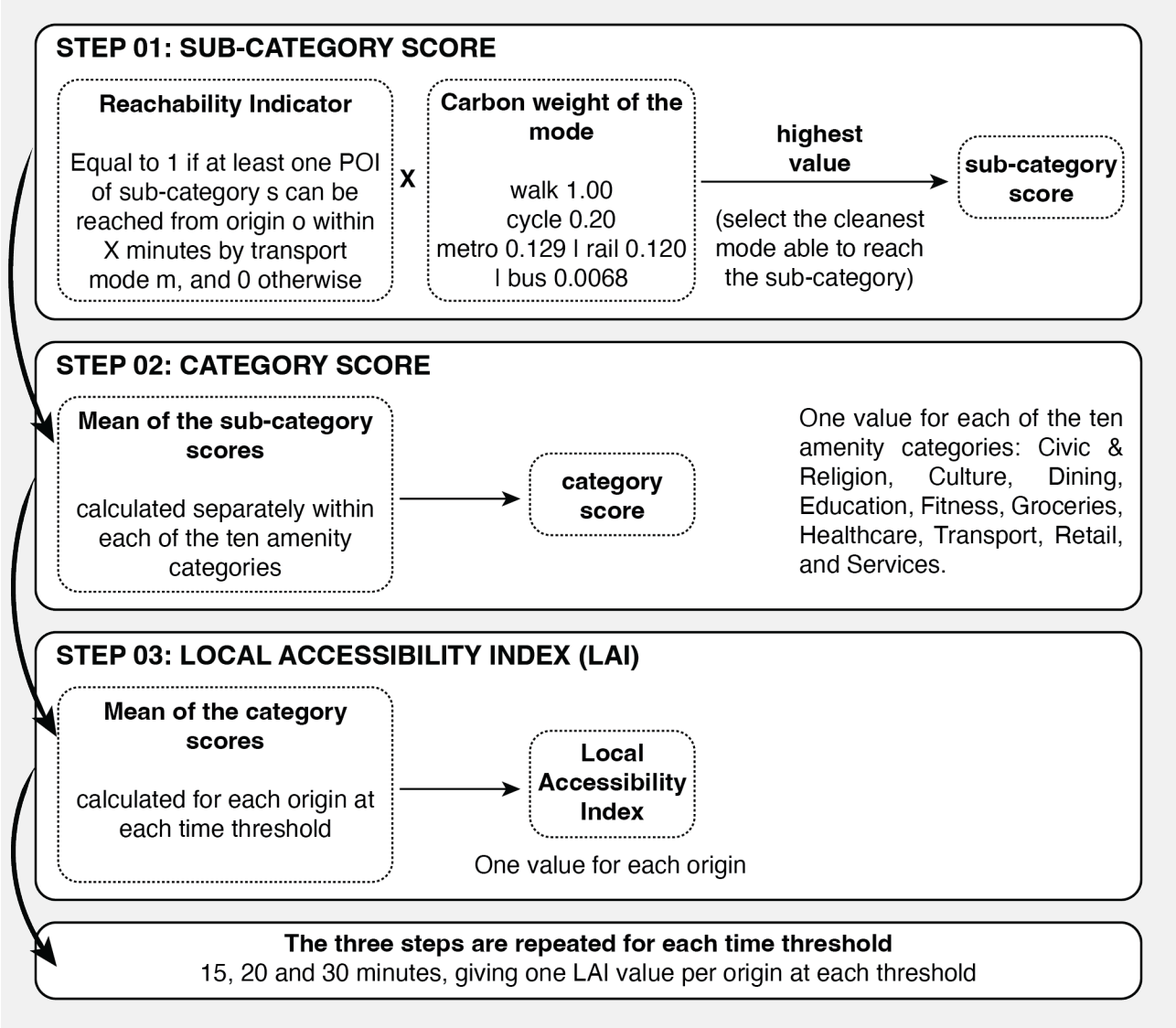
To assess how strongly the carbon weighting shapes the results, the index was also computed without it, as a control, assigning every transport mode an equal weight of 1.0. Comparing this control with the carbon-weighted index isolates the effect of the weighting on LAI scores.

[1] SPTrans data reported in Metrô-SP, 2022, and in CPTM, 2024.

4.3.5. Computing the index

The LAI is computed at three steps (Figure 06). First, the amenity sub-category score is calculated. Second, the amenity category score is calculated. Third, category scores are aggregated in the final LAI.

Figure 06: Index flowchart



Source: Author

Regarding amenity sub-category score, the first calculation determines the reachability indicator. This indicator records whether at least one POI of each sub-category can be reached, equal to 1 when it can and 0 when it cannot.

For an origin ***o***, a threshold ***X*** ∈ {15, 20, 30} minutes, and transport mode ***m*** ∈ ***M*** = {walk, cycle, bus, metro, rail} let $a^{X}_{o,s,m}$ be a reachability indicator of an amenity sub-category ***s***.

$$a^{X}_{o,s,m} = \begin{cases} 1, & \text{if at least one POI in sub-category } s \text{ is reachable from } o \text{ within } X \text{ minutes by mode } m, \\ 0, & \text{otherwise.} \end{cases}$$

The reachability indicator is computed for all transport modes, and it is multiplied by the carbon weight w_m of mode ***m***. Finally, the sub-category score $v^X o, s$ is the one from the lowest carbon emission mode able to reach it. For example, walk carries the maximum weight (1.00) and bus the minimum weight (0.0068), so an amenity sub-category reachable on foot scores 1.00, whereas one reachable only by bus scores 0.0068. Moreover, if the sub-category is reachable by walk and bus, the index retains only the walk score. Equation 01 describes the amenity sub-category score calculation.

Equation 01:

$$v_{o,s}^X = \underset{m \in M}{max}\left(w_m a_{o,s,m}^X\right)$$

The category score $C^X o, k$ is then calculated as the mean of the scores of all sub-categories $v^X o, s$ (even unreachable ones), where S_k is the set of sub-categories making up category k. Equation 02 describes the amenity category score calculation.

Equation 02:

$$C_{o,k}^X = \frac{1}{|S_k|} \sum_{s \in S_k} v_{o,s}^X$$

Finally, the Local Accessibility Index (LAI) $I^X o$ of the origin ***o*** is the mean of the ten K category scores $C^X o, k$. Equation 03 describes how LAI is calculated.

Equation 03:

$$I_o^X = \frac{1}{|K|} \sum_{k \in K} C_{o,k}^X, |K| = 10, I_o^X \in [0,1]$$

Equation Symbol Definitions:

o — origin (coordinate point);

X — time threshold (15, 20 or 30 min)

m — transport mode,

M = walk, cycle, bus, metro, rail

w_m — its carbon weight

s — amenity sub-category;

S_k — sub-categories belonging to category k

k — amenity category;

K — the ten categories ($|K| = 10$)

$a^X o, s, m$ — reachability indicator (1/0);

$v^X o, s$ — sub-category score;

$C^X o, k$ — category score;

$I^X o$ — LAI

A value close to 1 corresponds to an origin from which most types of daily amenity are within reach, while a value close to 0 corresponds to one from which few are. Because the outputs of each step are recorded, and each origin retains the economic class and expansion factor of the traveller who departed from it, the index can be disaggregated by time threshold, transport mode, amenity category and economic class, as set out in Section 4.5.

4.4. Hexagonal spatial representation

Once the LAI results were assigned to coordinate points, they were aggregated to easy visualisation at a scale less granular and suitable for local accessibility analysis, using Uber's H3 hierarchical indexing system. Each origin was assigned to the hexagonal cell that contains its coordinates. Where more than one origin fell within the same cell, the value of that cell was taken as the mean of the LAI values of those origins. Cells containing no origin were left empty. This was repeated for each time threshold and each amenity category.

Two resolutions were tested: level 10 (edge ~75 metres) and level 9 (edge ~175 metres). Level 9 was selected because each cell contains more sampled coordinate points, providing a better balance between spatial detail and a sufficient number of origins for meaningful analysis.

4.5. LAI interpretation

This study explored LAI results from two strategies to analyse inequalities in local accessibility. The first is a place-based strategy, which analyses the index at the level of the origins. The second is an individual-based strategy, which shifts the unit of analysis from the origin to the person.

4.5.1. Place-based analysis

The place-based analysis was done in four steps. First, the hexagonal spatial representation of LAI values was used to analyse how potential local accessibility is distributed across the territory, both for the overall index and for each amenity category.

Second, spatial clustering was measured with two techniques. Global Moran's I (Moran, 1950) measured how clustered each amenity category is across the city. Getis-Ord Gi* (Getis & Ord, 1992) was then applied, first to the overall index and then to each amenity category, to find high and low accessibility clusters. In each case, the index values were aggregated into H3 hexagons, and Getis-Ord Gi* was calculated using binary queen-contiguity weights, which compare each hexagon and its neighbours with the overall distribution. Each hexagon was then classified by its Getis-Ord Gi* z-score at the 95% confidence level as a hotspot ($z > 1.96$), where high accessibility clusters, or a coldspot ($z < -1.96$), where low accessibility clusters. The spatial analysis was repeated for the three time thresholds.

Third, to compare accessibility across economic classes, each origin was grouped by economic class of the person who departed from it, and LAI values were compared between classes using a

mean weighted by person expansion factor, so that results represent the whole population of São Paulo.

Fourth, to examine the contribution of each transport mode, three cumulative scenarios were compared: walking only; walking combined with cycling; and walking, cycling and public transport together. The modes are added in ascending order of carbon emissions so that the most sustainable modes are accounted for first and higher-emitting modes only afterwards. Because the scenarios are nested (each mode added on top of the previous one) the comparison isolates the marginal contribution of each mode, that is, whether introducing cycling, and subsequently public transport, allows an individual to reach amenity sub-categories and categories that were not already accessible through the preceding mode(s).

4.5.2. Individual-based analysis

To investigate accessibility from the individual-based strategy, the accessibility of the anchor (Section 4.1) was attributed to the individual carrying out the trip from that location. As individuals carry out activities departing from multiple anchors, three versions of the index are used: residential local accessibility, work/study local accessibility, and overall individual accessibility. The latter is the mean of LAI values for all anchor locations related to that individual. The comparison between these three versions indicates how individual potential local accessibility changes for the same person during the day.

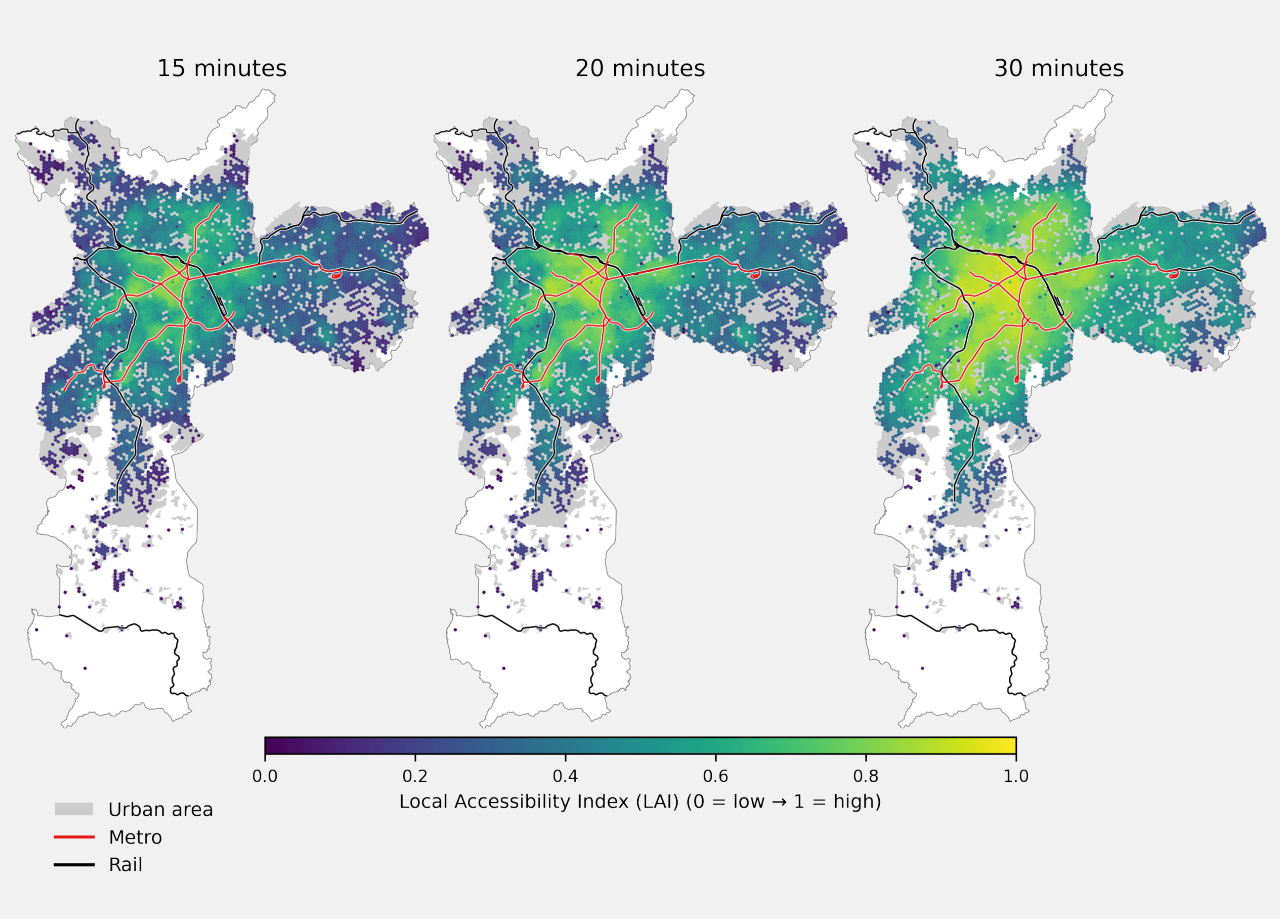
5. Results

5.1. Place-based analysis

5.1.1. Overall accessibility across São Paulo

Figure 07 shows the spatial distribution of the LAI across the municipality of SP, where a concentration of high values can be observed in the city centre, with high-value "corridors" also visible in other areas, most of which coincide with the metro and rail lines. The results show that LAI values increase as the time threshold increases, which is expected, since more time allows longer distances to be travelled to reach urban facilities and services.

Figure 07: LAI at X minutes



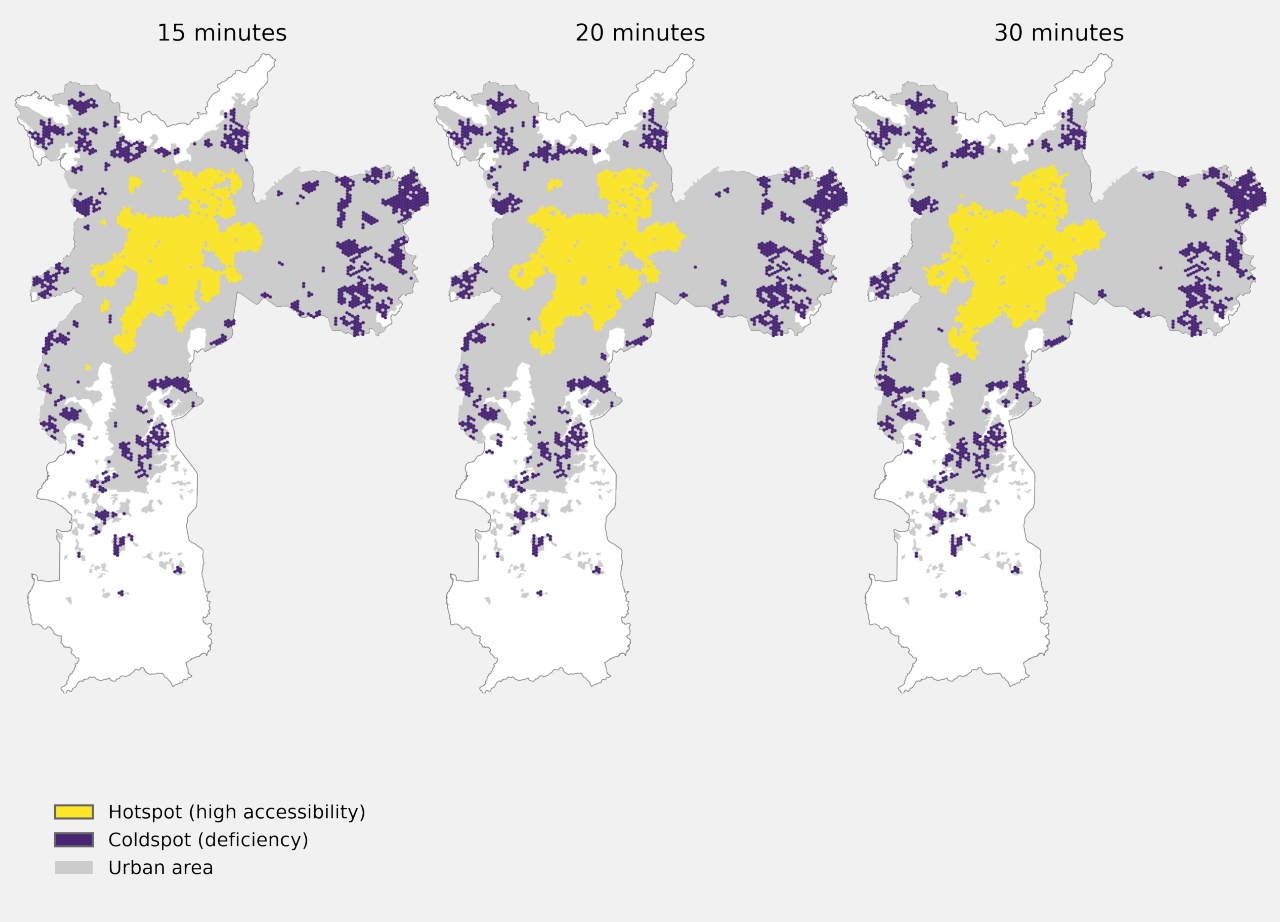
Source: Author

Within a 15-minute threshold, only the central area of the municipality of São Paulo and the areas adjacent to the metro lines show high accessibility indices. For a 20-minute threshold, the values of the indices in the centre and the areas adjacent to the metro lines increase, and the coverage area of these regions is also larger. However, even within a 20-minute threshold, the city's outskirts continue to have low accessibility indices. Within a 30-minute threshold, the same pattern of increased coverage in the city centre and the areas near the metro lines persists.

Conversely, within this threshold, there is a significant increase in accessibility indices in the outskirts. The mean index rises across the thresholds, from 0.51 at 15 minutes to 0.60 at 20 minutes and 0.71 at 30 minutes.

Figure 08 shows the Getis-Ord Gi* analysis of the overall LAI. It reveals accessibility hotspots (clusters of high values) in the city centre and coldspots (clusters of low values) in peripheral areas, especially in the eastern part of São Paulo. The pattern remains very stable across the three time thresholds.

Figure 08: LAI's Getis-Ord Gi* analysis at X minutes

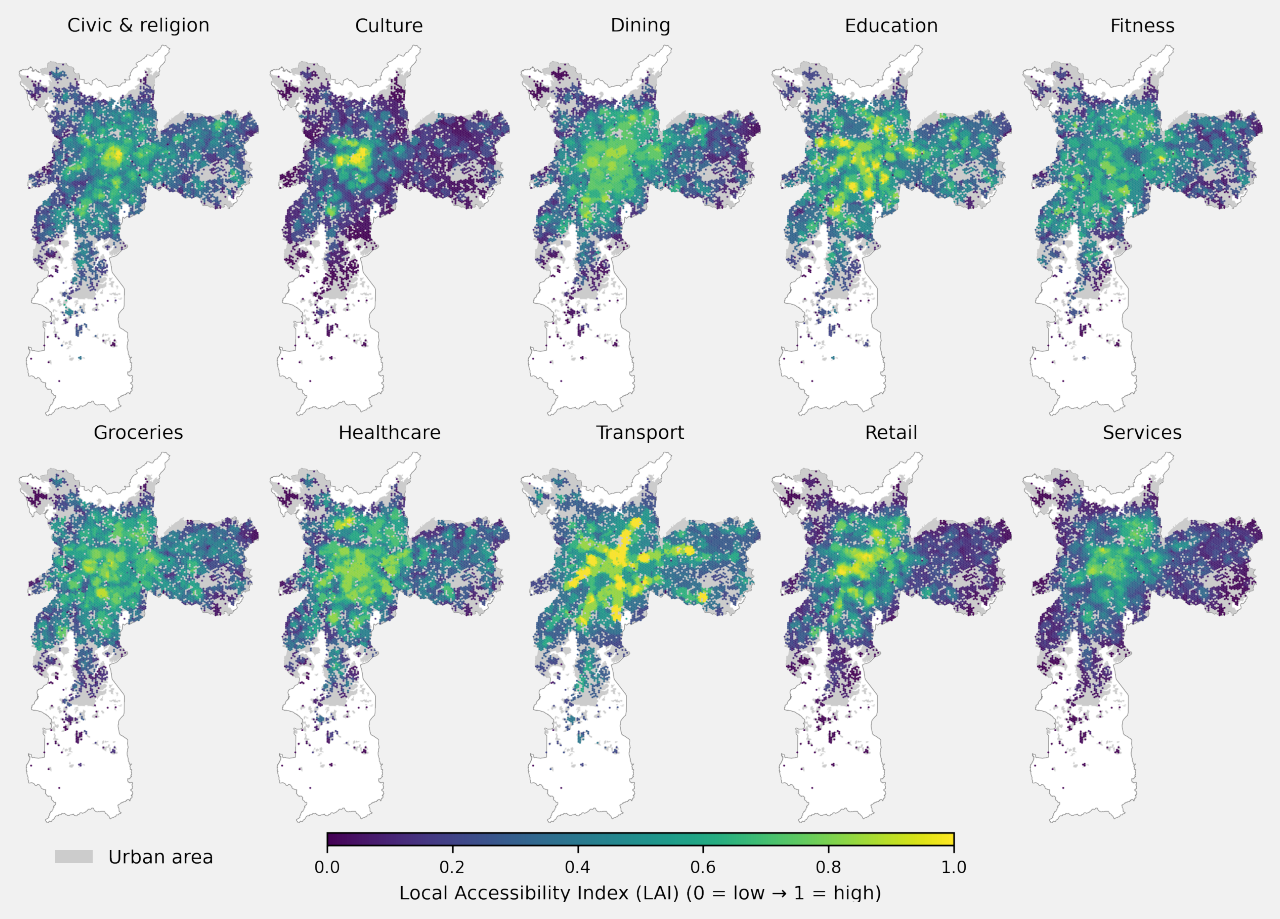


Source: Author

5.1.2. Accessibility by amenity category

Figure 09 shows the distribution of LAI per category at 15 minutes. Table 01 reports the Global Moran's I value for each category, all close to 1, confirming that every category is strongly clustered.

Figure 09: LAI per category at 15 minutes



Source: Author

Table 01: Global Moran's I values of the per-category LAI at 15 minutes

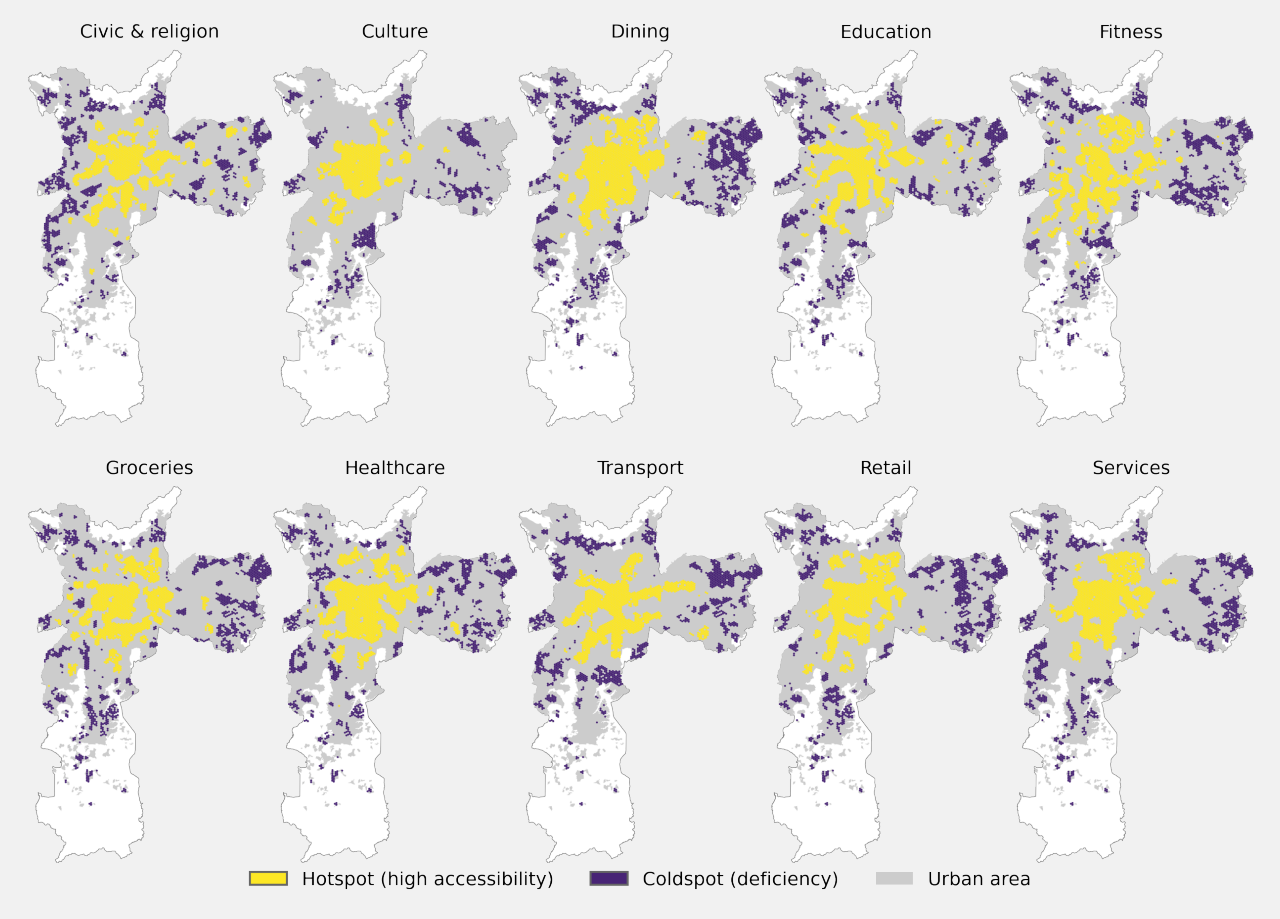
Global Moran's I value of the per-category LAI at 15 minutes									
Fitness	Education	Civic & Religion	Healthcare	Transport	Groceries	Dinning	Culture	Retail	Services
0.87	0.88	0.90	0.90	0.90	0.91	0.93	0.93	0.95	0.95

Source: Author

While Global Moran's I shows that every category is clustered, Figure 10 shows where those clusters are according to the Getis-Ord Gi* analysis. For all ten categories, the hotspots are concentrated in the central and central-western area of the municipality, whereas the coldspots lie in peripheral areas, especially in the far east.

It is important to consider that the far-eastern peripheral area of São Paulo is one of the metropolis's most economically disadvantaged areas, populated mainly by lower-income households, predominantly classes C to D–E (Lisboa, 2022). Some categories stand out: Culture has one of the smallest and most compact central hotspots, and transport produces a more elongated hotspot that follows the city's main metro axes rather than a single compact core.

Figure 10: LAI's Getis-Ord Gi* analysis at 15 minutes per category



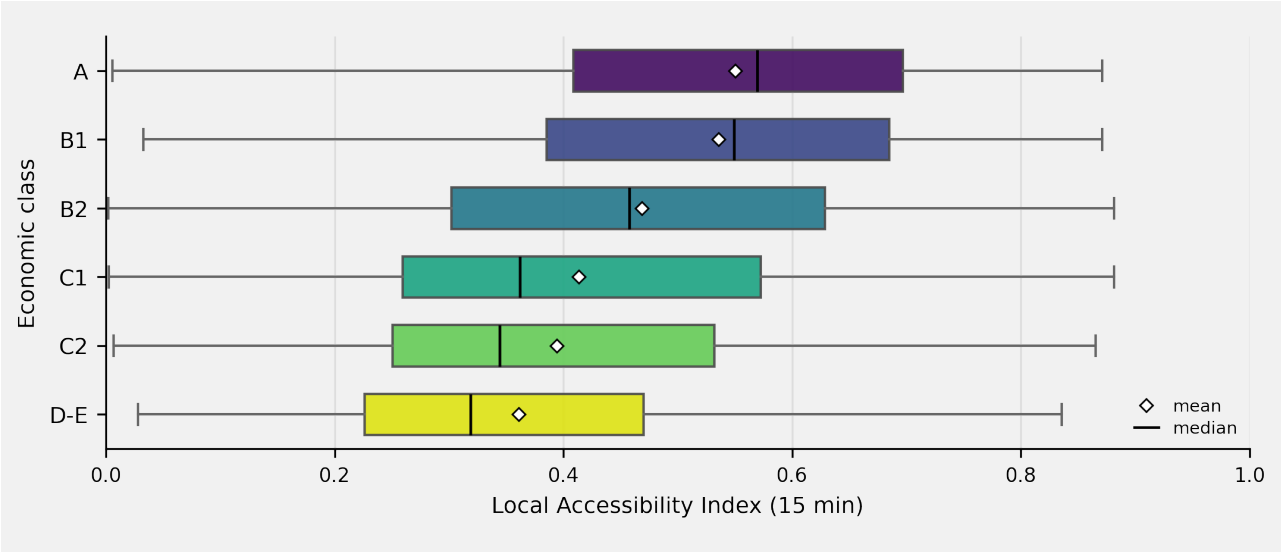
Source: Author

5.1.3. Accessibility across economic classes

Figure 11 shows how the index varies across economic classes. At 15 minutes, the mean index falls from 0.550 in class A to 0.361 in class D-E (B1 = 0.536, B2 = 0.468, C1 = 0.414, C2 = 0.394). This fall is not linear. Classes A and B1 are almost the same, with only 0.015 between them, and the three lowest classes are also close to each other. Most of the drop happens in the middle: two thirds of the difference between the six classes (0.122 of 0.189) is between B1 and C1. It separates a well-served group at the top from the rest of the city, and it shows little difference between the three poorest classes. In addition, the overlap between classes indicates that economic class explains only part of the variation in the index. Another relevant factor is the strong relationship between economic category and residential location in Brazilian society.

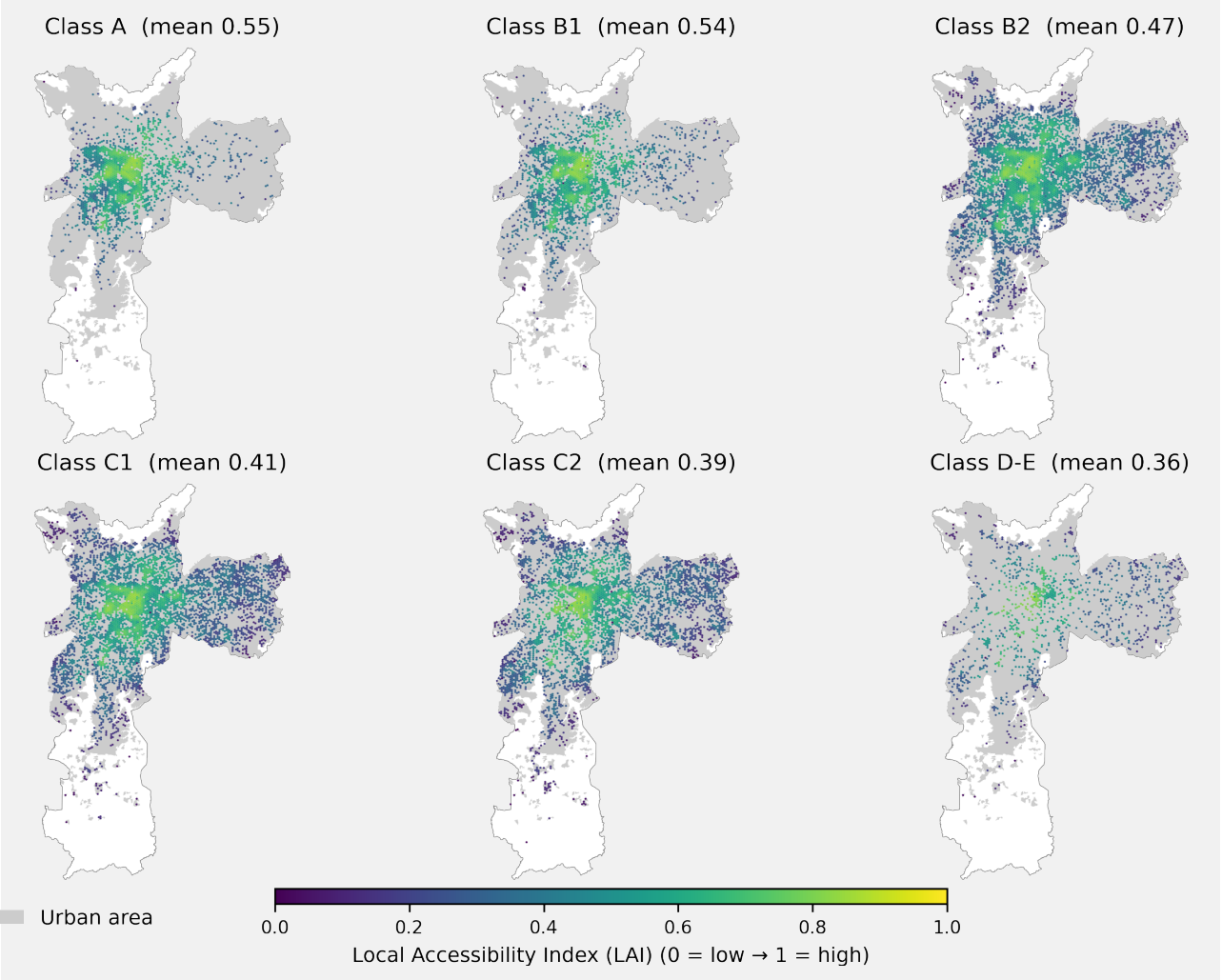
Figure 12 shows that inequality between groups is not merely numerical, but also spatial. Groups A and B1 predominantly travelled from origins in the centre of São Paulo, where accessibility levels are higher, while groups B2, C1, C2 and D-E had stronger presence in peripheral areas.

Figure 11: Boxplot LAI per economic class at 15 minutes



Source: Author

Figure 12: LAI per economic class at 15 minutes



Source: Author

5.1.4. Transport modes and carbon weight

Table 02 shows that adding cycling to walking raises the median LAI considerably across all thresholds, indicating that the bicycle extends access to amenity sub-categories and categories beyond those reachable on foot. By contrast, the further inclusion of public transport produces no change in the key statistics of the index at any threshold. This suggests that, within local accessibility context, public transport does not provide access to additional amenity types beyond those already reached by walking and cycling.

Table 02: Contribution of transport modes (carbon weighted)

Threshold	Transport mode	LAI			
		Mean	Median	25th Quartile	75th Quartile
15-Minutes	Walk	0.434	0.446	0.245	0.617
	Walk + Bike	0.505	0.526	0.329	0.677
	Walk + Bike + Public Transport	0.505	0.526	0.329	0.677
20-Minutes	Walk	0.532	0.561	0.337	0.730
	Walk + Bike	0.595	0.631	0.418	0.774
	Walk + Bike + Public Transport	0.595	0.631	0.418	0.774
30-Minutes	Walk	0.665	0.740	0.492	0.847
	Walk + Bike	0.714	0.784	0.564	0.872
	Walk + Bike + Public Transport	0.714	0.784	0.564	0.872

Source: Author

To assess if the results including public transport could be explained by the carbon weighting, the index was also computed without the weighting, giving every mode an equal weight of 1.0. Table 03 summarises its results. Removing the weighting raises the overall index substantially. For example, the mean 15-minute index rises from 0.505 to 0.788 for walking, cycling and public transport together. However, transport mode contribution pattern did not change. Public transport still adds no significative change into LAI at any threshold.

The weighting also affects the gap between economic classes. This gap can be seen by comparing the average 15-minute index of each class, in Table 04. Without weighting, the mean 15-minute index of the richest class (A) is about 1.28 times higher than that of the poorest class (D-E), and with weighting this difference grows to 1.52 times.

Table 03: Contribution of transport modes (without carbon weight)

Threshold	Transport mode	LAI			
		Mean	Median	25th Quartile	75th Quartile
15-Minutes	Walk	0.434	0.446	0.245	0.617
	Walk + Bike	0.787	0.865	0.676	0.929
	Walk + Bike + Public Transport	0.788	0.865	0.677	0.929
20-Minutes	Walk	0.532	0.561	0.337	0.730
	Walk + Bike	0.848	0.920	0.774	0.949
	Walk + Bike + Public Transport	0.849	0.920	0.775	0.949
30-Minutes	Walk	0.665	0.740	0.492	0.847
	Walk + Bike	0.908	0.957	0.893	0.969
	Walk + Bike + Public Transport	0.910	0.957	0.893	0.969

Source: Author

Table 04: Mean LAI per economic class with and without carbon weight

	Mean LAI (carbon weighted)			Mean LAI (carbon non-weighted)		
Classes	15 min	20 min	30 min	15 min	20 min	30 min
A	0.550	0.647	0.769	0.845	0.899	0.945
B1	0.536	0.629	0.751	0.829	0.886	0.937
B2	0.468	0.558	0.681	0.760	0.830	0.899
C1	0.414	0.500	0.622	0.704	0.782	0.867
C2	0.394	0.479	0.600	0.681	0.760	0.851
D–E	0.361	0.444	0.572	0.656	0.740	0.842

Source: Author

5.2. Individual-based analysis

Aiming to analyse accessibility not only from the origin/location perspective, but also from individual perspective, the index also looked at each individual filtered from 2023 SP OD survey, using three values: their overall index, the index at their home, and the index at their work/study place. Figure 13 shows that all three get lower from class A to class D-E, but the biggest difference is at home: the average home index falls from 0.50 (A) to 0.30 (D-E), while the work/study index falls less, from 0.60 (A) to 0.45 (D-E). Access is therefore most unequal where people live, and more equal where they work. The gap between work and home is also bigger for the lower-middle classes: for C1 and C2 the work index is about 0.16 higher than the home index, compared with about 0.10 for class A.

Figure 13: Individual LAI per economic class at 15 minutes



Source: Author

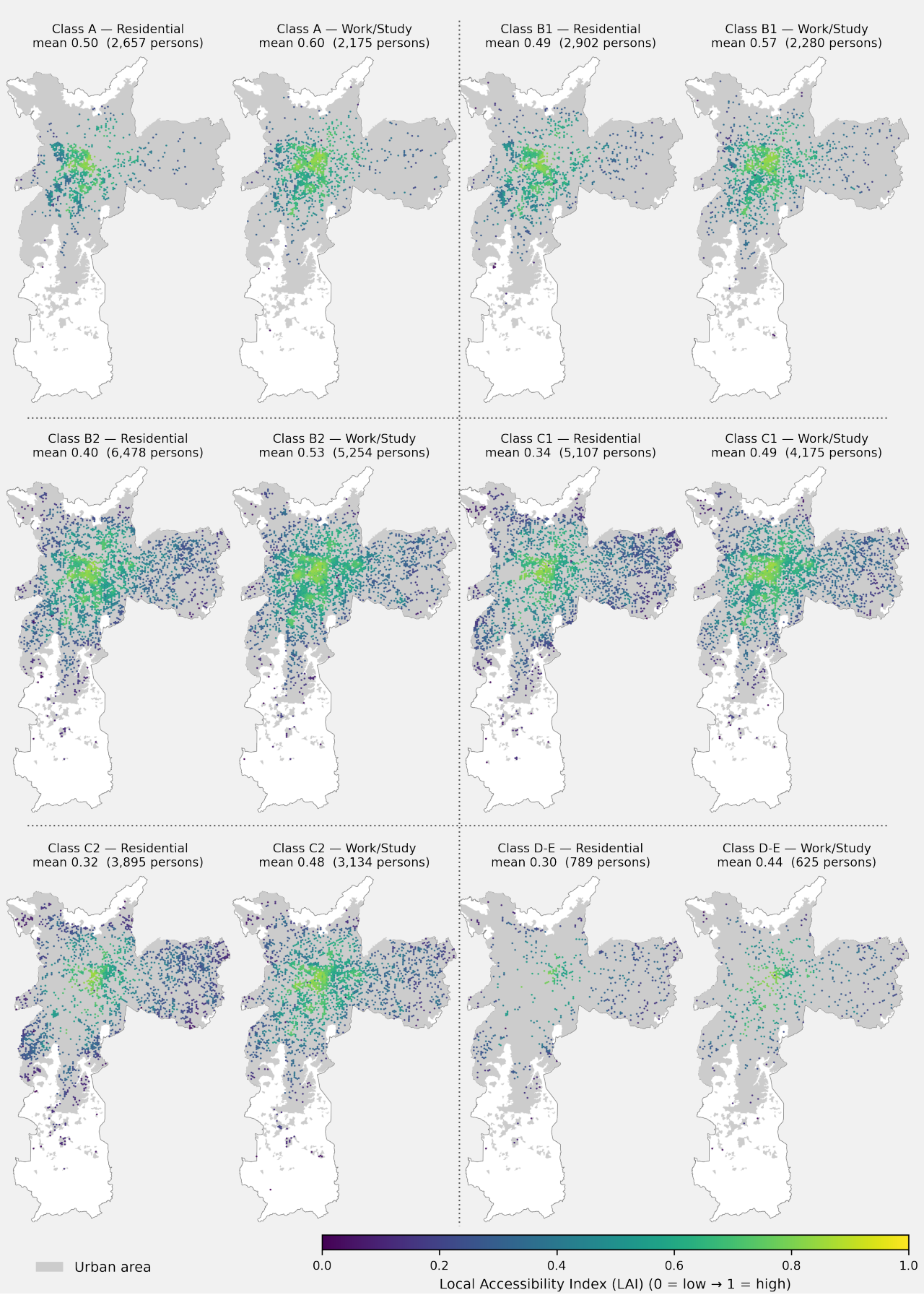
Figure 14 shows the LAI, at the 15-minute threshold, at the locations where individuals of each economic class live (residential origins) and work/study. It illustrates how the gap between residential and work/study accessibility shown in Figure 11 manifest it self in the territory.

Figure 15 adds a further dimension to the analysis: the population density attached to each origin, estimated with the survey expansion factors, and shown as the height of each hexagon. For this map only, the origins were grouped at H3 resolution 7 (edge ~1.2 km), because larger cells produce fewer and bigger bars, which makes the three-dimensional comparison easier to read.

Across all classes, work/study origins concentrate in the city centre, where the index is higher. Results also reveal that residential density shifts towards peripheral area as economic power decreases. Finally, classes D-E form the smallest group in this sample, although the economic class distribution published by ABEP[1] for the São Paulo metropolitan area indicates they should be about twice the size of class A (ABEP, 2024). This mismatch has at least three causes. First, the OD survey captures a smaller D-E share than ABEP even at the metropolitan scale (7.1% against 11.8%), reflecting differences between the two instruments. Second, D-E is the most metropolitan-peripheral class: half of its respondents (50.2%) live outside the municipality of São Paulo, so restricting the analysis to trips within the municipality removes more of this class than of any other. Third, D-E has the highest immobility rate: 48% of its members made no trip on the survey day (against 31% in class A), and a trip-based analysis cannot observe them properly.

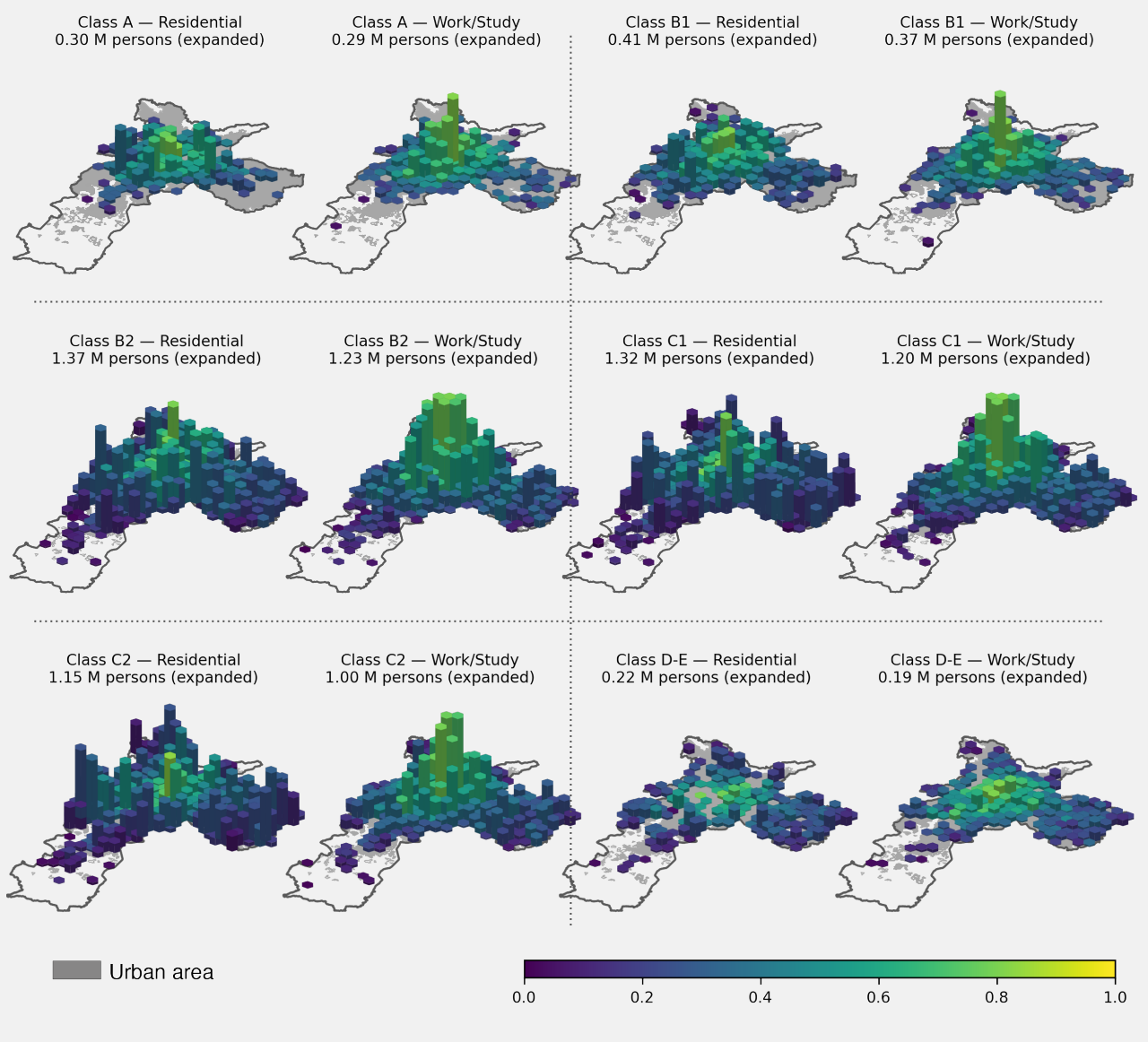
[1] Brazilian Association of Market Research Companies (translation made by the author)

Figure 14: Anchors' LAI per economic class at 15 minutes



Source: Author

Figure 15: Anchors' LAI and population density per economic class at 15 minutes



Source: Author

6. Discussion

Results point to two main findings. First, potential local accessibility to the XmC concentrates in the centre of São Paulo and along the metro and rail corridors, and is much lower in peripheral areas. Second, accessibility reduces as economic power reduces. Both patterns are consistent with accessibility research for São Paulo (Vieira and Haddad, 2015; Slovic et al., 2019; Boisjoly et al., 2020; Saraiva and Barros, 2022), what is new is that these patterns remain the same at the local scale, for smaller thresholds. The following sections discuss where proximity is found, how it is delivered, at what carbon cost, and for whom it is a real possibility.

6.1. Proximity spatial distribution

The concentration of hotspots in the central and central-western area, and coldspots in peripheral areas (Sections 5.1.1 and 5.1.2), matches the way São Paulo is organised. Amenities and jobs are concentrated in the central and central-western area, while the East Zone is a lower-income, mostly residential peripheral area with fewer amenities (Villaça, 2011; Feitosa et al., 2021). It also matches the pattern found in other Global South cities (Section 3.2), where proximity tends to concentrate where wealth concentrates, as Israel et al. (2026) found in Santiago. The results also connect to the Strategic Master Plan (PDE), which aims to bring housing and services together along public transport corridors. This is already the case in the central and central-western area, where transport corridors, amenities and jobs are in the same place and the index is highest. It is not the case in peripheral areas, where corridors are few, amenities are scarce and the index is low.

However, the alignment between hight LAI and rail corridors reveals another paradox. As this study looks at potential local accessibility over short time thresholds, where most trips would be made on foot or by bike, and public transport barely changes the index, it is hard to attribute the high potential local accessibility to the metro and rail presence. Another explanation would be that São Paulo's rail network runs through the consolidated, high-density central and central-western core, which is precisely where amenities and jobs concentrate (Villaça, 2011), so it is the land-use component, the density and diversity of nearby amenities (Geurs and van Wee, 2004), that drives high LAI values in São Paulo. This also explains why local accessibility is low at peripheral areas, especially in the far east. These are poorer, low-density areas that are mostly residential (Villaça, 2011; Lisboa, 2022), and offer fewer amenities within walking or cycling range. The missing metro lines there is probably a result of the same lack of investment, and not the cause of the low local accessibility. As a consequence, a large part of the city is still far from the proximity ideal.

Results also show that local accessibility increases as the time threshold increases, but this increase is not the same for everyone or everywhere. The central and central-western areas already reach a high level of accessibility at 15 minutes, so they gain little from a longer threshold. Peripheral areas, and the lower-income groups who live there, start from a much lower level and gain a bit more as the threshold grows. However, lower classes stay below the richer classes at every threshold (Table 04).

6.2. Reaching amenities: walking, cycling and public transport

A further question concerns which travel modes enable potential local accessibility. Several studies argue that public transport greatly expands accessibility (Calafiore et al., 2022; Birkenfeld et al., 2024; Mouratidis, 2024). In São Paulo, however, adding public transport to walking and cycling changed the index almost nothing, at the local threshold used.

The unweighted version of the index showed that the reason is not the carbon weighting. Even without the carbon penalty, public transport adds little to the cumulative local accessibility, because cycling already reaches almost everything that public transport could reach within the same time threshold. Within 15 minutes a bicycle covers about 3.75 km, and within 30 minutes about 7.5 km, with no waiting time. A public transport journey, by contrast, spends much of the time budget walking to the stop and waiting for the service. Cycling therefore does most of the work of extending local accessibility beyond walking, while public transport adds little on top of it.

This points to a second, less expected result. Cycling expands potential accessibility considerably, so it could play a large role in local accessibility across São Paulo. Yet few people cycle. Only 1.32% of all trips in the 2023 SP OD survey were made by bicycle (Metrô-SP, 2023). Even for trips within cycling range, the share stays very low. Among trips up to 3.75 km, which a bicycle covers in about 15 minutes, only 1.7% are made by bicycle while 46% are made on foot. Among trips up to 7.5 km, the 30-minute range, the bicycle share is 1.6% and walking is 38% (Metrô-SP, 2023). People walk these short trips far more than they cycle them, even though the bicycle could cover longer distances. Network coverage alone does not explain, or induce, bicycle use.

The literature and the São Paulo context point to conditions that may discourage cycling (Knebel et al., 2024). Some concern whether a person is able to cycle at all: owning a bicycle (Florindo et al., 2018; Kienteka et al., 2014; Ogilvie et al., 2008; Stahley et al., 2022) and knowing how to ride. Safety is a further barrier, both from sharing the road with motor traffic (Knebel et al., 2024; Heesch et al., 2014; Pearson et al., 2023; Russell et al., 2021; Twaddle et al., 2010) and from the risk of theft, including while parked. Another barrier relates to the mobility of care: the everyday trips people make to look after others, such as taking children to school. These trips are harder to make by bicycle (Montoya-Robledo et al., 2020).

Overall, the low use of the bicycle in São Paulo is not explained by how far it can reach. Even the short trips are rarely made by bicycle, and are walked instead. The low use is better explained by the conditions that make cycling difficult. This gap between the access the bicycle could provide and how rarely it is chosen is itself one of the findings of this study.

6.3. Carbon emissions associated to proximity

The carbon weighting changes the gap between economic classes much more than it changes the accessibility map. Without weighting, the mean 15-minute LAI of the richest class (A) is about 1.28 times higher than that of the poorest (D-E). With weighting, this difference grows to 1.52

times. This happens because opportunities are not equally distributed across the city. People in richer, central areas can reach many amenities on foot, and walking has the highest weight. People in poorer outer areas depend more on cycling and public transport, which have lower weights. As a consequence, low-carbon potential accessibility in São Paulo was, in 2023, a real possibility mostly for the higher economic classes. The poorest groups can reach amenities mostly through the modes that pollute more, and yet it is the higher income groups that use the car the most.

This finding also shows why proximity by itself will not make the 15mC green for everyone. Walking is the cleanest way to reach amenities, but only people who live close to them can walk. The poorest groups live far from most amenities, so their access to greener mobility will depend on having cleaner public transport modes and more amenities in the outer areas.

6.4. For whom proximity is possible

Classes D-E, C2, and C1 have less economic power, travel longer distances every day, and have less free time due to their long commutes. Because they have the fewest ways to reach amenities far from home, they depend more on local access, yet they are the ones who live furthest from where potential local accessibility is higher. This is consistent with a pattern in which lower-income groups reach fewer opportunities than wealthier ones, both in Latin American cities (Guzman et al., 2024; Israel et al., 2026) and in São Paulo (Slovic et al., 2019; Bittencourt et al., 2021; Feitosa et al., 2021; Lisboa, 2022; Saraiva and Barros, 2022). This inequality also has a spatial side. The individual-based analysis showed that work/study places are more central and better served, so access there is higher across classes. It is in the residential areas that the gap between classes opens up.

This matches a concern already raised in the literature. Israel et al. (2026), Guzman et al. (2024) and Abbiasov et al. (2024) argue that, in segregated cities, promoting local accessibility can make existing inequality worse, because it mostly benefits the areas that already have amenities, while the areas with few amenities gain little from it. Policies such as the Strategic Master Plan (PDE) illustrate this scenario, once it aims bring housing and services together along public transport corridors, regions in which LAI is already high. If the aim is to reduce inequality and promote accessibility for all, investment in local accessibility should go to the places where LAI is currently lowest.

However, bringing amenities to deprived areas does not guarantee equity promotion in the medium and long term. Better-serviced areas tend to become more expensive to live in, so 15mC policies risk leading to gentrification when applied to deprived areas (Elldér, 2024), a process that already exists for the housing sector in Latin American cities (Barros, 2004). Additionally, diversity of services needs consumers that can afford to sustain it. Avoiding gentrification therefore requires policies that take this economic constraint into account and allow low-income residents to remain in the areas where XmC initiatives are applied.

6.5. Strengths and limitations

Unlike most applications of the 15-minute city, the proposed index is built from the origins where people actually start their trips on a typical weekday, using the 2023 SP OD survey. It brings together, in a single measure, elements that the studies reviewed in section 3.2 kept separate: three time thresholds, several transport modes, and a carbon weighting, applied across the whole municipality. The results are also broken down by economic class, and an individual-based approach makes it possible to compare how potential local accessibility changes over the day for the same individuals. A control version of the index, without the carbon weighting, shows how much the weighting affects the results.

At the same time, the study carries limitations. Some of them are related to datasets. Once this study is analysing local accessibility, an important limitation is that 2023 SP OD Survey excluded walking trips shorter than 500 metres, unless the trip purpose was work or study. Regarding amenity data, OSM is a crowdsourced dataset, mapped by volunteers, so its coverage is uneven and tends to be more complete in central and wealthier areas. Moreover, SPTrans only makes the most recent version of its public transport schedules (GTFS) available for download. The routing therefore used the 2026 schedules, while the survey data is from 2023, so the two do not correspond exactly.

Other limitations regard methodology. The index counts amenities by presence and sub-category diversity, but treats every amenity of the same sub-category as equal, regardless of its size, capacity or quality. This matters because an accessibility measure is meaningful if it reflects what residents actually value (Handy and Niemeier, 1997), and being close to a service does not guarantee that the service is adequate. The same point was raised for the Global South by Nieto-Combariza et al. (2026). In addition, physical proximity does not mean that individuals can actually take advantage of the opportunities. Affordability and cultural fit are also relevant factors to define who could access an amenity. Additionally, the carbon weighting relies on life-cycle multipliers from Chester and Horvath (2009), a study based on the United States. Applying these values to São Paulo is an approximation, because the infrastructure and vehicles they measure differ between the two countries. Finally, the index measures potential accessibility (what can be reached) and not whether people actually use the amenities near them.

7. Conclusion

Regarding the first research question, the study has shown that from selected origin points, potential local accessibility is unevenly distributed throughout the city, with higher levels in the city centre and along rail corridors, and lower levels in peripheral areas, especially the eastern part of São Paulo. In addition, cycling extends access to amenity sub-categories and categories beyond those reachable on foot and has a bigger impact than public transport in the cumulative low-carbon emission analysis done here. Finally, potential local accessibility increases as the time threshold increases, but not equally over the territory. A longer time budget helps the less-served areas the most, but it does not close the gap between central and peripheral areas. The centre already meets the proximity ideal at 15 minutes, while peripheral areas reach a similar level only with 30 minutes, and often not even then.

Concerning the second research question, results indicate that selected origins visited by higher economic classes have higher potential accessibility. Individual-based analysis also indicates that this gap is larger at people's homes than at their work or study places, which shows that the inequality between economic classes comes mainly from where people live rather than from where they work.

This study addresses the gap identified in the literature in two ways. The first contribution is methodological. It proposes a data-based index that measures potential local accessibility from the origins where people travel on a typical weekday, and brings together time thresholds, transport modes, a carbon weighting and results by economic class in a single measure.

The second contribution is empirical. Studies using thresholds of 60 to 90 minutes have already shown that accessibility in São Paulo is higher for higher-income groups and lower as distance from the centre increases (Section 2.1). This study complement previous findings by showing that the same pattern remains at 15 to 30 minutes, so proximity reproduces the wider pattern rather than offering an alternative to it. Furthermore, at a local scale, the inclusion of public transport adds only a small number of amenities and sub-categories beyond those already reachable on foot and by bicycle. This differs from studies using higher thresholds, where public transport is found to expand accessibility considerably (Section 4.1). Finally, the carbon weighting shows that low-carbon proximity is concentrated in the wealthier central areas, a dimension no previous accessibility study of São Paulo has measured.

The results suggest some directions for policy. In peripheral areas, disadvantaged groups potentially reach more amenities in longer thresholds, indicating that they would need to travel longer distances. In this scenario, faster transport modes, such as cycling and public transport, would have a greater role to deliver local accessibility. A direction is to expand cycling infrastructure, since cycling is the mode that most extends local access beyond walking. Beyond infrastructure, encouraging cycling will also require more research on how to make it more affordable, safe and inclusive. Another alternative is to invest in low-carbon public transport, since peripheral areas rely heavily on buses, which are the most polluting of these modes. One

example is the electric bus fleet that the city has already started to introduce (Prefeitura de São Paulo, 2023). Finally, this study could be repeated as new editions of the Origin Destination Survey are released, providing a tool to track local accessibility overtime in São Paulo.

The methodological and empirical contributions of this study also open directions for future research. Perhaps the most important is to include the quality of amenities in the index, and not only their presence. This could be done with both quantitative and qualitative methods, and by taking into account how residents themselves value the services near them (section 6.5). A second direction is to compare potential accessibility with real travel behaviour, to see how far what can be reached matches what people actually do. Future research could also replace the carbon life-cycle values used here, which come from the United States, with values specific to São Paulo or Brazil. Adding longer thresholds, such as 60 and 90 minutes, would extend the analysis to longer journeys and to the metropolitan scale. Finally, applying this methodology to other Global South cities would make it possible to compare results, identify shared patterns, and look for common solutions for improving proximity in these contexts.

This brings the study back to the paradox in the title. The XmC is presented as a model that can bring proximity to everyone, but in São Paulo proximity already exists mostly in the wealthy centre and along the transport corridors, and it is missing where the poorest residents live. These are the people who most depend on local access, because they have the fewest ways and time budgets to reach amenities far from home. In a city as unequal as São Paulo, applying the model in the same way everywhere does not close this gap, and can even make it wider. For the Global South, this shows that the XmC cannot be treated as a one-size-fits-all solution. Proximity is still a goal worth pursuing, but reaching it fairly means recognising that different parts of a city start from very different conditions. A tool like the LAI can help, by showing where proximity already exists, for whom, and where it is most needed, so that the concept can be adapted to each city instead of applied as a ready-made model.

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Appendix A. Supplementary data

Table 05: Modal share of trips, 2023 SP OD Survey (weighted by the trip expansion factor)

Code	Mode	%
1	Metro	7.66%
2	Train	3.03%
3	Monorail	0.12%
4	Bus – São Paulo city	11.49%
5	Bus – other municipalities	2.94%
6	Bus – metropolitan	1.44%
7	Chartered	0.90%
8	School transport	6.83%
9	Car driver	21.39%
10	Car passenger	7.92%
11	Taxi	0.19%
12	App taxi	2.92%
13	Moto driver	3.16%
14	Moto passenger	0.28%
15	Mototaxi passenger	0.02%
16	Bicycle	1.32%
17	Walk	28.2%
18	Other	0.19%
	Active Mobility	29.52%
	Motor Vehicles	70.29%
	Public Transport	26.68%
	Private Transport	43.61%
	Others	0.19%
	Total	**100%**

Source: Author (adapted from 2023 SP OD Survey)

Table 06: Transport mode share within each economic class, 2023 SP OD Survey

Class	Walk	Cycle	Private car	Public transport	Motorcycle	Taxi/ App	Chartered/ School	Other	Total
A	11.31	0.58	66.43	9.36	2.03	4.12	6.10	0.07	100.00
B1	16.24	1.14	53.79	15.42	2.94	3.79	6.64	0.03	100.00
B2	20.81	0.75	40.58	23.66	3.88	3.57	6.52	0.24	100.00
C1	28.73	1.17	25.30	29.28	4.14	2.76	8.49	0.13	100.00
C2	38.82	1.83	11.05	33.14	3.27	2.68	8.99	0.23	100.00
D–E	46.63	3.54	6.52	31.95	1.11	2.73	7.05	0.47	100.00

Source: Author (adapted from 2023 SP OD Survey)

Table 07: Amenity categories and sub-categories OSM key and values

Amenity Categories	OSM key	Amenity Sub-categories OSM values (key=value)
Civic & Religion	amenity	place_of_worship, townhall, courthouse, police, fire_station, post_office, community_centre, social_facility, public_building
Culture	amenity	theatre, cinema, arts_centre, museum, library
	tourism	museum, gallery
Dining	amenity	restaurant, fast_food, cafe, bar, pub, food_court, biergarten
Education	amenity	school, university, college, kindergarten, language_school
Fitness	amenity	gym, sports_centre
	leisure	fitness_centre, sports_centre, swimming_pool, track
	sport	swimming
Groceries	shop	supermarket, convenience, grocery, greengrocer, butcher, bakery, deli, health_food
Healthcare	amenity	hospital, clinic, doctors, dentist, pharmacy
	healthcare	hospital, clinic, centre
Transport	amenity	bus_station, taxi
	public_transport	station
	railway	station, subway_entrance
	highway	bus_stop

Amenity Categories	OSM key	Amenity Sub-categories OSM values (key=value)
Retail	shop	mall, department_store, clothes, electronics, shoes, general, hardware, books, toys, furniture, pet, florist, stationery, optician, jewelry
Services	shop	hairdresser, beauty, dry_cleaning, laundry, tailor
	amenity	bank, atm, bureau_de_change, car_wash, charging_station, state_agent, car_rental
	office	insurance, lawyer

Source: Author

Appendix B. Data availability

Access the GitHub repository for this dissertation.

Appendix C. Research Log

Table 08 shows the research log for the supervision meetings done during this dissertation, emails are not included.

Table 08: Research Log

	Supervision Meetings	
	Date	**Discussed Subject**
1	11/03/2026	Discussing initial concepts for my dissertation: 15-Minute City, Paradox (Zang et al., 2025), Daytime socio-spatial diversity.
2	01/04/2026	Talked about ethics (Form A: Screening), data, some new references in segregation studies (Lisboa, 2022).
3	06/05/2026	Discussed exploratory data analysis, literature reviewed so far, and the calendar/agenda for future steps.
4	20/05/2026	Discussed some concepts, such as ecological fallacy, Latin America growth and development, accessibility, segregation. Also discussed research question formulation.
5	05/06/2026	Discussed chaining trips concept, data wrangling and next steps.
6	09/07/2026	Feedback on first written chapters (methodology and results) and guide lines for next chapters to write.
7	31/07/2026	Discussed place-based and individual-based analysis, and feedback on introduction and literature review chapters.
8	04/08/2026	Introduction workshop
9	05/08/2026	Feedback on introduction and discussion about chained trips' application.

Source: Author